A Control Matrix for Scientific Variable-Role Probing

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Abstract

Interpretability requires experiments that distinguish a model mechanism from properties of the measurement. We examine this problem in linear probes for five variable roles in code models. The study includes three models and draws from all seven programming languages in XLCoST, although coverage is not a complete model-by-language grid. High probe accuracy supports different conclusions under matched controls. Misleading names reduce index probe F1 by 0.272 yet increase accumulator F1 by 0.079; iterator probes can be refit after the same intervention in two models. On Python, boolean occurrence type is decodable at 0.981–0.988 macro-F1, but a masked source-line classifier reaches 0.983 without a language model. Rerunning both predictors with aligned predictions gives paired problem-clustered deltas of -0.003 to $+0.002$ across the three models, with intervals excluding any probe advantage above 0.014–0.017: a resolved negative. Class or structure labels remain decodable after perturbation-specific refitting and pass a within-program label-permutation control in three models, yet a gate-specified activation patch at the query-name site fails in Qwen2.5-1.5B. These outcomes motivate a control matrix: decodability, cue sensitivity, surface sufficiency, and site-specific causal relevance are distinct claims. Recording the unit, estimand, and unresolved alternatives makes the evidence more falsifiable and replicable.

1 Introduction

Linear probes are often used to ask what information a representation contains. A successful probe establishes decodability under a particular readout. It does not by itself establish that the model computed an abstract feature, that the feature is unavailable in surface form, or that the model uses it behaviorally [Belinkov, 2022, Voita and Titov, 2020]. Control tasks constrain probe capacity [Hewitt and Liang, 2019], and targeted studies show that probe accuracy need not entail task relevance [Ravichander et al., 2021, Elazar et al., 2021]. Capacity is only one source of ambiguity. Input leakage, label construction, sample resolution, and an unmatched intervention can each preserve high accuracy while changing the scientific claim.

This distinction is central to interpretability as a science. A measurement is useful when its target is explicit, its alternative explanations are tested, and contrary outcomes can change the conclusion. We use variable-role probing as a comparative case study. Variable roles describe how an identifier functions in a program, such as accumulating values or indexing a container [Sajaniemi, 2002]. They are attractive probe targets because the same nominal role can be expressed by names, syntax, and distributed context. That multiplicity also makes probe accuracy underidentified. Prior code-probing studies, including the 15-task INSPECT framework, map decodable syntactic and semantic information across layers and models [Karmakar and Robbes, 2021, Troshin and Chirkova, 2022, Karmakar and Robbes, 2024]. We do not propose another taxonomy. We contribute a role-by-control evidence map that records which interpretation survives, which comparison is incommensurate, and what outcome would falsify each surviving claim.

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Table 1: Actual evidence coverage. No row should be read as a complete three-model by seven-language experiment.

Role	Models	Language coverage	Strongest available controls
Accumulator	Qwen2.5-1.5B	Python plus Java, C++, C, C#	Six name interventions, cross-language transfer
Index or key	Qwen2.5-1.5B	Python, Java, JavaScript, C++, C, C#	Misleading and collapsed names, cross-language transfer
Iterator	Coder-1.5B, StarCoder2-7B	Python, Java, JavaScript, PHP, C++, C, C#	Six name interventions, cross-language transfer
Boolean	All three	Python, Java, JavaScript, PHP	Five refit renames, surface baselines, name-label control task
Class or struct	All three	Python, C++, JavaScript, C	Six refit renames, within-program label permutation, causal patching

Our evidence covers five roles, three models, and the seven-language XLCoST corpus [Zhu et al., 2022]. The experiment is intentionally reported as an uneven evidence matrix rather than a full factorial study. Qwen2.5-1.5B, Qwen2.5-Coder-1.5B, and StarCoder2-7B appear in the controlled boolean and class-reference experiments [Qwen et al., 2024, Hui et al., 2024, Lozhkov et al., 2024]. Earlier accumulator and index runs use one model and a weaker split protocol. Iterator runs cover two models and all seven languages. This is a methodological case study of evidential boundaries, not a validation of the matrix as a universal framework. Its heterogeneity separates replicated findings from leads that still require confirmation.

The central result is not that one representation type wins. It is that the same headline observation, high linear-probe F1, decomposes into at least four outcomes. Index probes depend strongly on names. Accumulator and iterator probes can be refit after misleading renames. A model-free classifier nearly matches boolean probe scores, with no retained paired uncertainty for the difference. Class-reference probes survive tested renames and a randomized- label control but fail a gate-specified causal intervention at one site. A single accuracy number cannot identify among these explanations.

2 Study design and evidence matrix

Roles, models, and languages. We treat five roles as related case studies, not measurement-equivalent instances of one construct. Accumulators update within loops; indices or keys appear as loop counters or subscripts; iterators are loop-bound names and their uses; boolean labels distinguish assignment, conditional, loop, and return uses; class/structure labels mark declared type names and references. Binary token probes label all other tokens negative. Heuristic AST and regular-expression extractors vary by language and lack a common multilingual gold set. Protocols also differ: accumulator, index, and iterator use legacy token-stratified splits and test-maximized layers; boolean mean-pools occurrences, groups splits by problem, and predicts usage type; class/structure uses token labels, problem groups, and five seeds. The modern pipelines select layers on validation data. XLCoST contains C, C++, C#, Java, JavaScript, PHP, and Python. Table 1 states actual coverage; scores across rows do not estimate one common task.

Four control questions. Renaming asks whether lexical identity carries the prediction. We compare original names with numeric, single-character, collapsed, random-noun, and misleading names drawn from another role. Each condition refits its probe and selects its layer, so preserved F1 means that a similarly predictive readout can be recovered after renaming; it does not show that one fixed feature direction is invariant. A model-free character classifier with the target name masked asks whether local source text already determines the label. Randomized-label controls ask whether the probe family can fit a specified null task. Their nulls differ: boolean assigns a random label per variable name, while class/structure permutes labels within programs. Activation patching asks whether replacing a residual state at a gate-specified site moves a behavioral logit readout [Meng et al., 2022].

These controls are not interchangeable. Renaming can show name robustness while leaving syntax untouched; with refitting, it measures recoverability rather than invariance. A positive selectivity score

Table 2: Probe results after the strongest available control. Values are macro-F1. Deltas compare the misleading-name condition with original names.

Role	Baseline probe	Key control result	Strongest supported inference
Index	0.959	$\Delta = -0.272$	Refit performance depends on names in the legacy Qwen run
Accumulator	0.893	$\Delta = +0.079$	Refit performance survives that rename in the legacy run
Iterator	0.988, 0.990	$\Delta = +0.012, +0.009$	Refit performance survives misleading names in two models
Boolean	0.981–0.988	paired $\Delta \in [-0.003, +0.002]$	Probe advantage over masked local source excluded above 0.017
Class or struct	0.979–0.982	$\lvert\Delta\rvert \leq 0.004$	Refit robustness plus within-program permutation selectivity

can reject its specified null task while leaving a surface shortcut intact. A model-free comparator can establish surface sufficiency but cannot show how the model uses the feature. A causal intervention can ground a use claim only at its specified layer, span, direction, and readout.

Evidence grades. Boolean and class-reference experiments use problem-grouped splits, validation-selected layers, five seeds, and paired or clustered uncertainty. Iterator artifacts use two models but retain the legacy token-level split and test-maximized layer, without matched surface or capacity controls. Accumulator and index likewise use token-level splits, test-maximized layers, and whole-word renaming. We retain their large, opposite effects as a falsifiable hypothesis, not as a confirmed estimate of a population difference. A DeepSeek index run is excluded because a tokenizer offset error invalidates its activation alignment.

3 The same probe score supports different claims

3.1 Lexical intervention separates roles

Perturbation-specific refitting produces opposite effects for index and accumulator probes. The Qwen2.5-1.5B index probe falls from 0.959 to 0.687, a change of -0.272. The accumulator probe rises from 0.893 to 0.972, a change of $+0.079$. The intervention therefore does more than lower overall data quality: it changes how recoverable the two labels are under the legacy readout. This is the study's clearest dissociation, but occurrence-level splitting, test-maximized layers, and refitting prevent a fixed-direction invariance interpretation. Its numerical magnitude is exploratory.

Iterator provides a separate refit-robust result at near-ceiling baseline accuracy. Qwen2.5-Coder-1.5B and StarCoder2-7B reach baseline F1 of 0.988 and 0.990. Misleading iterator names change those scores by $+0.012$ and $+0.009$. Python-trained probes transfer across the other six XLCoST languages with F1 ranges of 0.936–0.983 and 0.906–0.964. These results support recoverability after the tested naming intervention. They do not establish feature invariance or abstraction because the probe and best layer are refit and the iterator runs lack a matched model-free surface comparator.

3.2 A model-free baseline resolves the boolean result

Boolean occurrence type illustrates why a high, replicated probe score can remain scientifically weak. On a frozen Python sample, the three models reach 0.982, 0.981, and 0.988 macro-F1. Selectivity ranges from $+0.272$ to $+0.288$, which rejects the specific hypothesis that an equally expressive probe predicts labels randomly assigned per variable name as well as the real usage labels. Yet a character classifier on the enclosing source line, with the variable name masked, reaches 0.983. The probe-minus-surface differences are -0.001, -0.002, and $+0.005$.

We reran both predictors on the frozen sample with aligned predictions kept (the original exports had none): the paired, problem-clustered probe-minus-surface deltas are -0.002 $[-0.017, 0.014]$, -0.003 $[-0.037, 0.014]$, and $+0.002$ $[-0.016, 0.017]$ over 915 problem clusters, the reruns reproducing the committed scores before pairing. Every interval covers zero, and each excludes a probe advantage larger than 0.014–0.017 macro-F1. Five renaming conditions also change refit performance by at

most 0.013. The supported claim is now sharper: boolean occurrence type is decodable after the tested renames, and the paired intervals rule out a meaningful probe advantage over masked local syntax on Python. The cross-language runs extend decodability to Java, JavaScript, and PHP. Only Python has renaming runs, so they do not extend the refit-robustness claim.

3.3 A causal null limits the class-reference claim

Class-reference probes pass two controls that boolean does not. Across all three models, baseline F1 is 0.979–0.982 and within-program permutation selectivity is $+0.503$–$+0.553$. Misleading class-style names change F1 by -0.004, -0.004, and $+0.004$. Python-trained probes also transfer to C++, JavaScript, and C at 0.885–0.980. This supports a permutation-selective signal whose predictive performance can be recovered after the tested renames. It does not establish a fixed rename-invariant direction.

We then test a stronger causal claim in Qwen2.5-1.5B. The experiment contains 288 matched class-versus-function prompts in 48 template clusters. Let $D = \ell(\texttt{True}) - \ell(\texttt{False})$. Denoising patches the class-source residual into a function prompt and defines $e_d = D_{\text{patched function}} - D_{\text{function}}$; noising reverses the patch and defines $e_n = D_{\text{class}} - D_{\text{patched class}}$. Recovery is the ratio of the mean signed effect to the mean matched gap $D_{\text{class}} - D_{\text{function}}$. Percentile intervals resample template clusters. The gate additionally requires positive, control-separated effects, recovery of at least 0.05, and mean effects of at least 0.10 in both directions.

The behavioral readout has limited dynamic range. Both prompt classes prefer `True`. Their mean logit differences are $+2.75$ for class prompts and $+1.78$ for function prompts, although the matched class-minus-function gap is $+0.97$ and has the expected sign for 98.6% of pairs. The gate therefore tests whether patching transfers this relative distinction, not whether it flips a well-calibrated binary decision.

Denoising produces a mean effect of 0.0087 with a 95% interval $[-0.0007, 0.0188]$ and recovery 0.0089. Noising produces 0.0199 $[0.0109, 0.0296]$ with recovery 0.0204. The intervention therefore fails the specified gate. The protocol and results entered the repository together, so we do not claim auditable preregistration. This is a site-specific causal null, not evidence that class information is absent or unused everywhere. More directly, the intervals exclude mean effects above 0.0188 and 0.0296 logit-difference units in the two tested directions. Strong decodability, positive selectivity, and renaming robustness therefore do not establish a substantive bidirectional contribution at this site and readout.

4 A claim–control matrix for scientific probing

The experiments suggest orthogonal claims rather than a cumulative ladder.

1. **Decodability.** A held-out probe exceeds a simple label baseline.
2. **Probe-capacity control.** The probe exceeds a matched randomized-label task whose construction must be reported. This does not imply training dependence or surface irreducibility.
3. **Cue sensitivity.** A fixed probe evaluated on a paired rename isolates one question; perturbation-specific refitting instead measures recoverability after the cue changes.
4. **Surface sufficiency.** A model-free comparator on the same population tests whether the proposed surface information already predicts the label. A representational advantage additionally requires uncertainty on the paired performance difference.
5. **Site-specific causal relevance.** A gate-specified intervention changes a behavioral readout in both directions and separates from placebo and random controls.

No row entails the next. Boolean passes its name-label control and shows refit robustness, while masked local syntax remains sufficient. Class/structure passes a different permutation control, lacks a matched surface comparator, and fails the bidirectional causal gate at the selected site. Iterator shows refit robustness but lacks surface and capacity comparisons. Accumulator and index motivate a cue-type hypothesis but require modern replication before direct comparison.

Table 3: Scientific status of the paper's main interpretations. "Open" means that the required comparison has not been run under a matched protocol.

Interpretation	Evidence in this study	Status and decisive next test
Legacy roles may use different cues	Opposite refit deltas in incommensurate pipelines, plus iterator stability	Exploratory. Repeat with problem groups, scope-aware renaming, and intervals
Boolean exceeds Python local syntax	Paired reruns, three models, 915 problem clusters	Resolved negative: intervals exclude advantages above 0.014–0.017
Class refit performance survives renaming	Three-model permutation selectivity and misleading-name deltas near zero	Supported as recoverability. Evaluate one fixed probe across paired renames
Query-name signal makes a substantive bidirectional causal contribution	Two-direction patching with placebo and random controls	Not supported at Qwen layer 18. Run an explicitly exploratory all-layer sweep
Cross-language role abstraction	Strong transfer with uneven role and model coverage	Open. Apply one extractor and control battery to a complete language grid

Falsifying outcomes. The framework is useful only if results can change the interpretation. A problem-grouped rerun could shrink or reverse the accumulator-index dissociation. An all-layer class-reference patch could reveal a causal site outside the probe-selected layer. A scope-correct seven-language rename pipeline could expose language-specific cue dependence hidden by the current coverage. Each outcome would revise a named claim rather than leave an elastic assertion of representation.

Replication standard. We recommend publishing the evidence matrix with the metric table. Every cell should identify the prediction unit and label, model revision, corpus slice, split unit, layer-selection rule, random seeds, comparator input, uncertainty estimand, and intervention gate. Negative results should state the smallest effect the design can exclude. This turns a probing result from a single score into a set of auditable contrasts.

5 Identifiability and evaluation design

A useful contrast changes one named factor and carries uncertainty for the resulting difference. The boolean probes and masked-line classifier share a population, and the paired rerun turns their comparison into an interval: no model's probe advantage exceeds 0.017, which converts an unresolvable aggregate gap of 0.005 into a bounded negative. Renaming reuses surviving occurrences but refits probes and layers, so its delta measures recoverability after intervention, not fixed-direction invariance. The class experiment instead uses matched prompts plus placebo, random, and same-source controls. Its effects fall below the specified 0.10 gate, although noising has a positive interval; the substantive bidirectional claim, rather than all causal involvement, is rejected at that site.

6 Limitations

The study covers three models, five roles, and all seven XLCoST languages collectively, but not a complete $3 \times 5 \times 7$ design: boolean and class-reference are three-model, iterator two, accumulator and index one, and language coverage and extractors differ by role.

Protocol quality is uneven. The accumulator-index dissociation predates problem-grouped splits and scope-aware renaming; function overlap, textual replacement, test-maximized layers, and refitting may inflate it. Iterator lacks matched surface and control-task baselines. Boolean's local labels are nearly surface-predictable, and the paired rerun bounds any probe advantage below 0.017. Class-reference patching covers one model, layer, token site, and readout; stopping after the Qwen null was not registered in advance.

Except for the reported patch, probes remain correlational; mean pooling exposes identifiers, refitting tests recoverability rather than invariance, renames cover selected alternatives, and cross-language transfer may reflect shared syntax, names, or extractor artifacts. Hence: a matrix, not a conclusion that models represent abstract roles.

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A Per-role control results

Boolean probe figures. The baseline panels retain the individual surface comparators hidden by the best-baseline table. Renaming is available only for Python; the absence of corresponding figures in other languages is a coverage hole.

Table 4: Complete boolean occurrence-type probe and model-free baseline results. The best baseline is selected from name-only, masked statement, masked line, masked window, covariate-only, and majority features. ρ is a descriptive one-instance score step, not an interval or equivalence test; aligned probe/baseline predictions were not retained.

language	model	problems	probe F1	best baseline	difference	ρ	selectivity
Java	Qwen2.5-1.5B	1,427	0.982	0.980	+0.002	0.0120	+0.255
Java	Qwen2.5-Coder-1.5B	1,427	0.976	0.980	-0.004	0.0079	+0.271
Java	StarCoder2-7B	1,427	0.990	0.980	+0.010	0.0120	+0.243
JavaScript	Qwen2.5-1.5B	1,351	0.989	0.992	-0.003	0.0170	+0.132
JavaScript	Qwen2.5-Coder-1.5B	1,351	0.998	0.992	+0.006	0.0170	+0.150
JavaScript	StarCoder2-7B	1,351	0.988	0.992	-0.004	0.0170	+0.120
PHP	Qwen2.5-1.5B	1,305	0.983	0.969	+0.014	0.0080	+0.214
PHP	Qwen2.5-Coder-1.5B	1,305	0.975	0.969	+0.006	0.0080	+0.216
PHP	StarCoder2-7B	1,305	0.990	0.969	+0.021	0.0080	+0.240
Python	Qwen2.5-1.5B	1,301	0.982	0.983	-0.001	0.0068	+0.280
Python	Qwen2.5-Coder-1.5B	1,301	0.981	0.983	-0.002	0.0068	+0.272
Python	StarCoder2-7B	1,301	0.988	0.983	+0.005	0.0068	+0.288

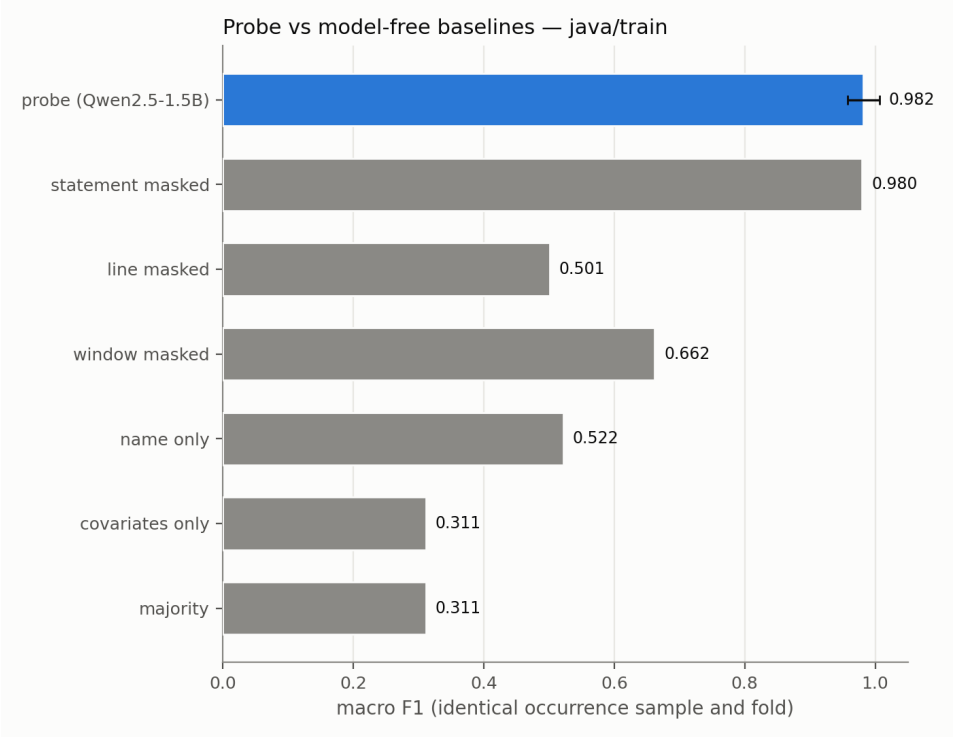

Figure 1: Boolean probe and model-free baselines for Java, Qwen2.5-1.5B.

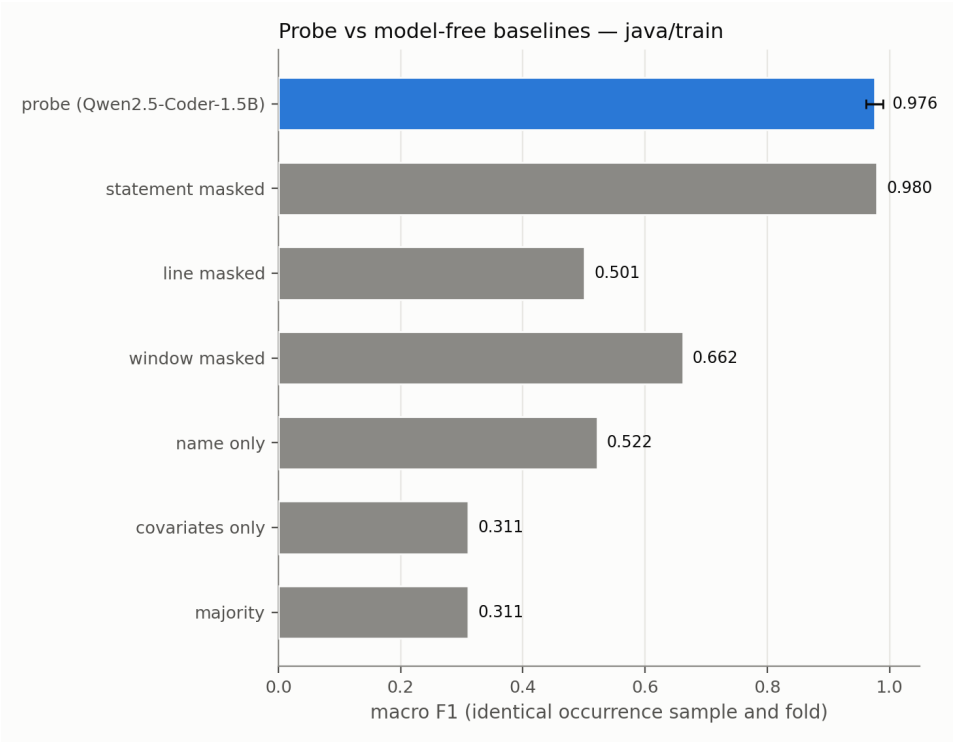

Figure 2: Boolean probe and model-free baselines for Java, Qwen2.5-Coder-1.5B.

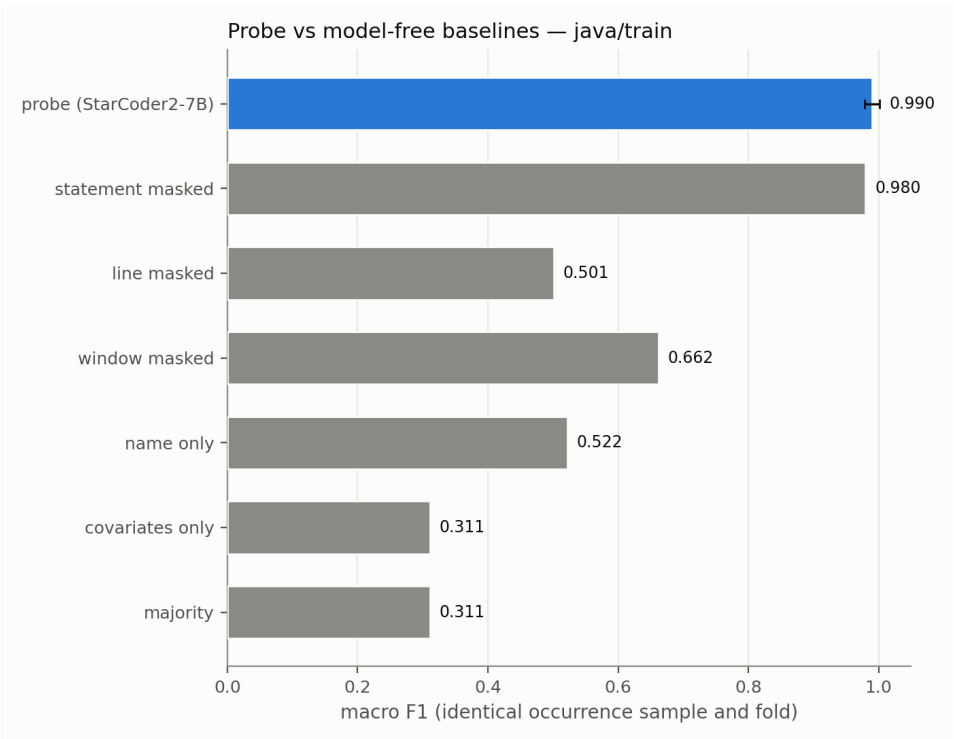

Figure 3: Boolean probe and model-free baselines for Java, StarCoder2-7B.

Table 5: Boolean Python identifier-renaming audit. C1–C5 are the five committed renaming conditions. Each condition refits a probe and reselects its layer, so deltas measure recoverability rather than fixed-direction invariance; intervals are problem-clustered. Empty renaming fields for other languages indicate that those runs were not committed, not a null result.

model	C1	C2	C3	C4	C5
Qwen2.5-1.5B	+0.003	-0.004	+0.001	-0.003	+0.001
Qwen2.5-Coder-1.5B	+0.001	-0.010	+0.006	-0.004	-0.006
StarCoder2-7B	-0.003	-0.013	-0.006	-0.001	-0.006

Table 6: Complete model-free boolean baselines by language. Baselines are shared across the three neural models because they use the same frozen examples and surface features.

language	majority	name	statement	line	window	covariates
Java	0.311	0.522	0.980	0.501	0.662	0.311
JavaScript	0.227	0.373	0.992	0.369	0.552	0.226
PHP	0.228	0.505	0.969	0.413	0.581	0.228
Python	0.217	0.422	0.970	0.983	0.605	0.216

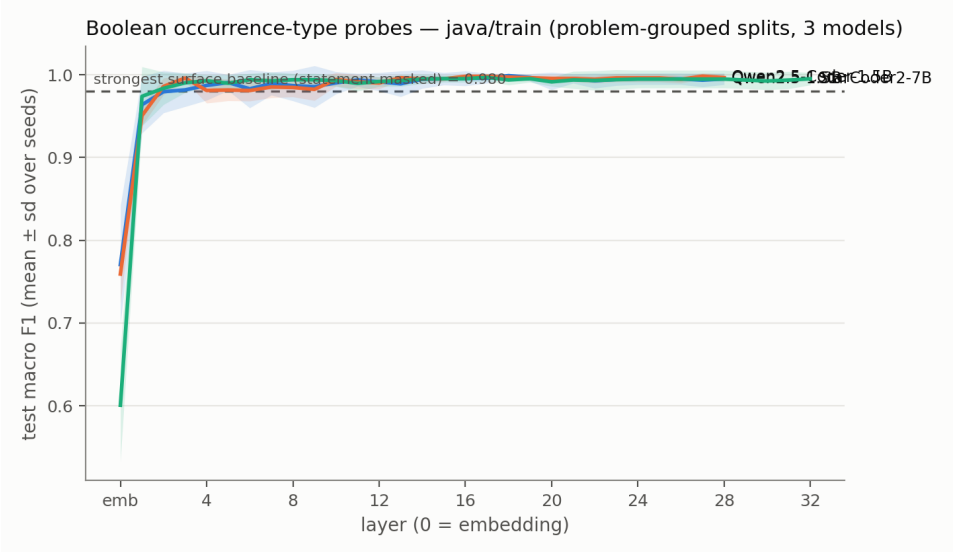

Figure 4: Boolean probe layer curves for Java across the three models.

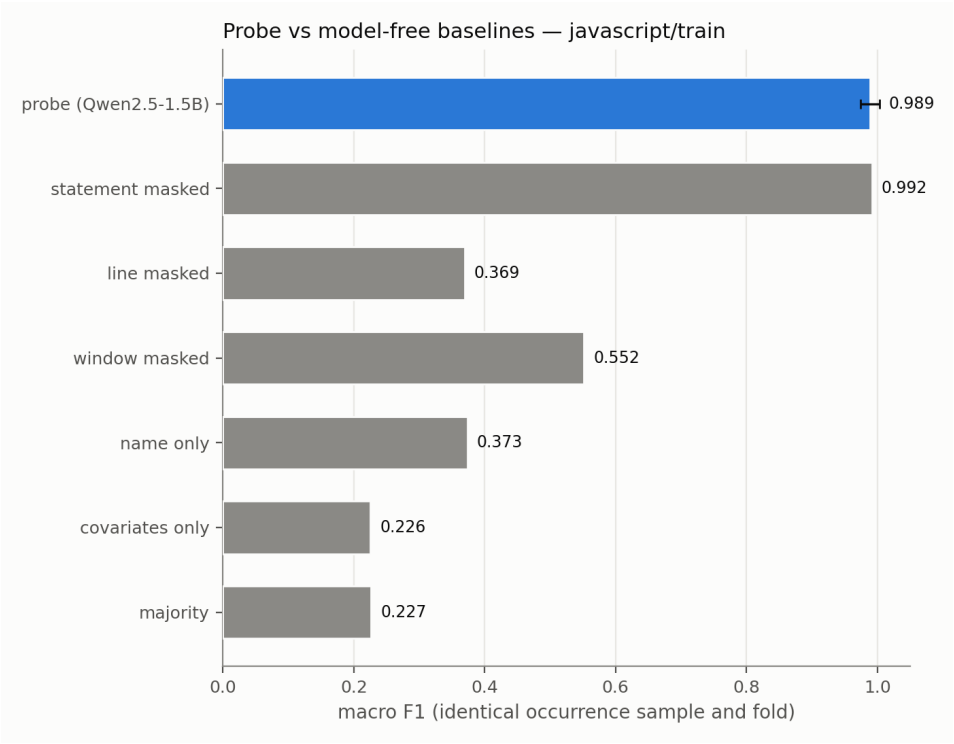

Figure 5: Boolean probe and model-free baselines for JavaScript, Qwen2.5-1.5B.

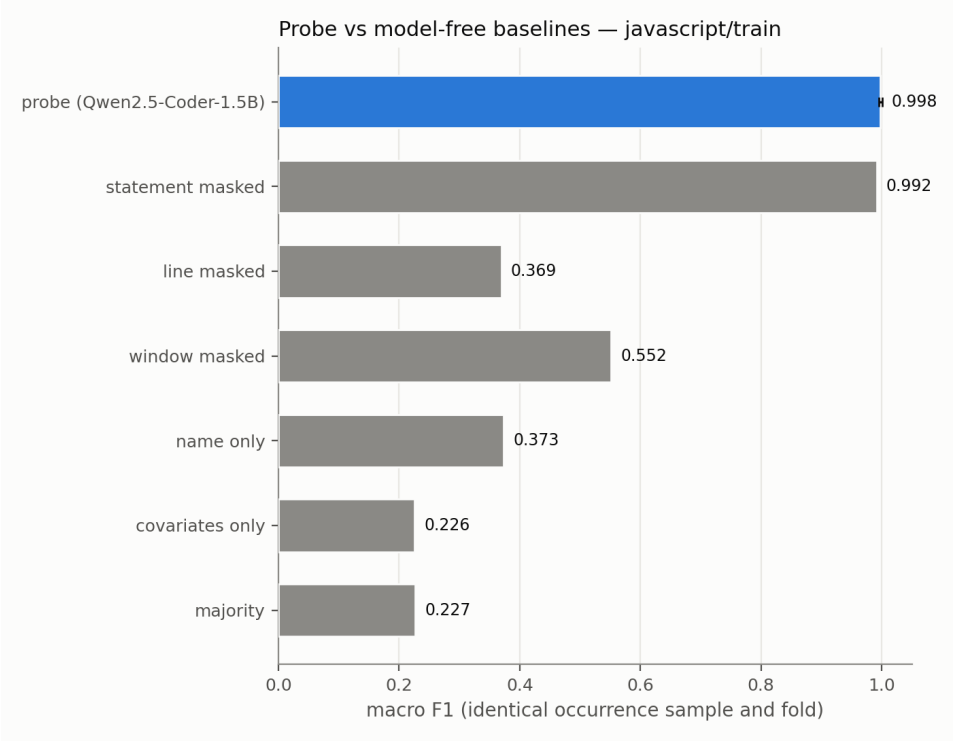

Figure 6: Boolean probe and model-free baselines for JavaScript, Qwen2.5-Coder-1.5B.

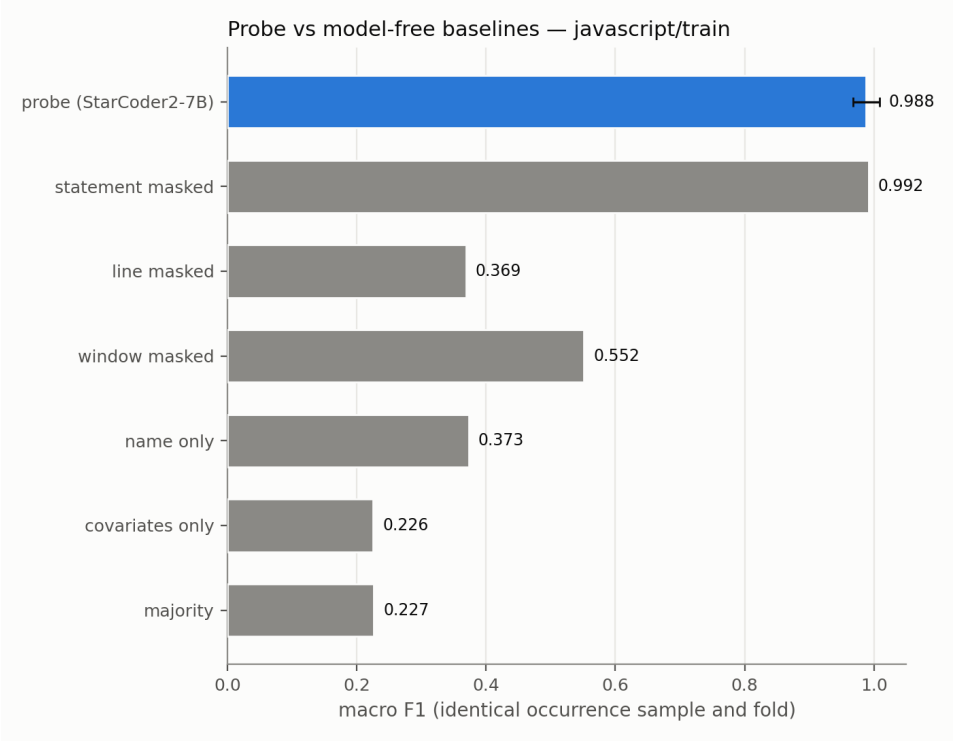

Figure 7: Boolean probe and model-free baselines for JavaScript, StarCoder2-7B.

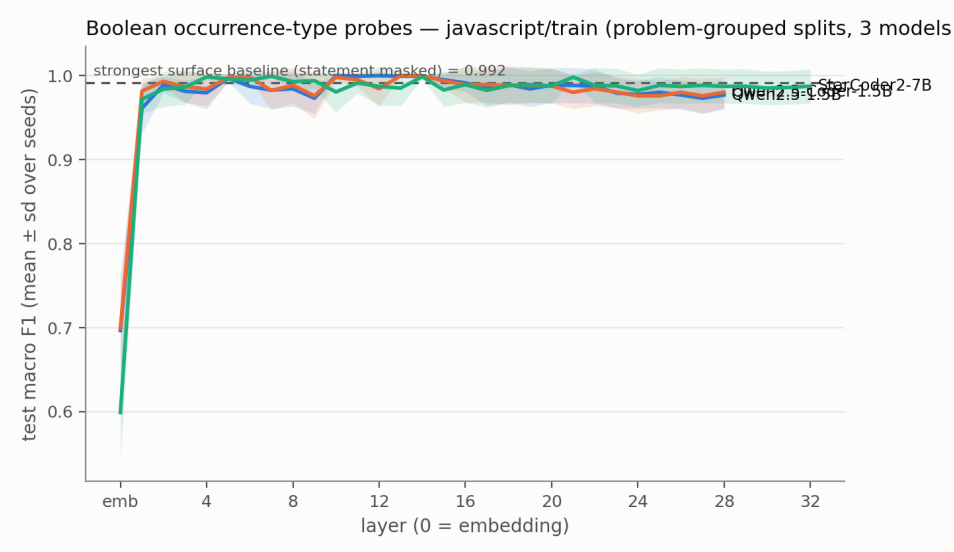

Figure 8: Boolean probe layer curves for JavaScript across the three models.

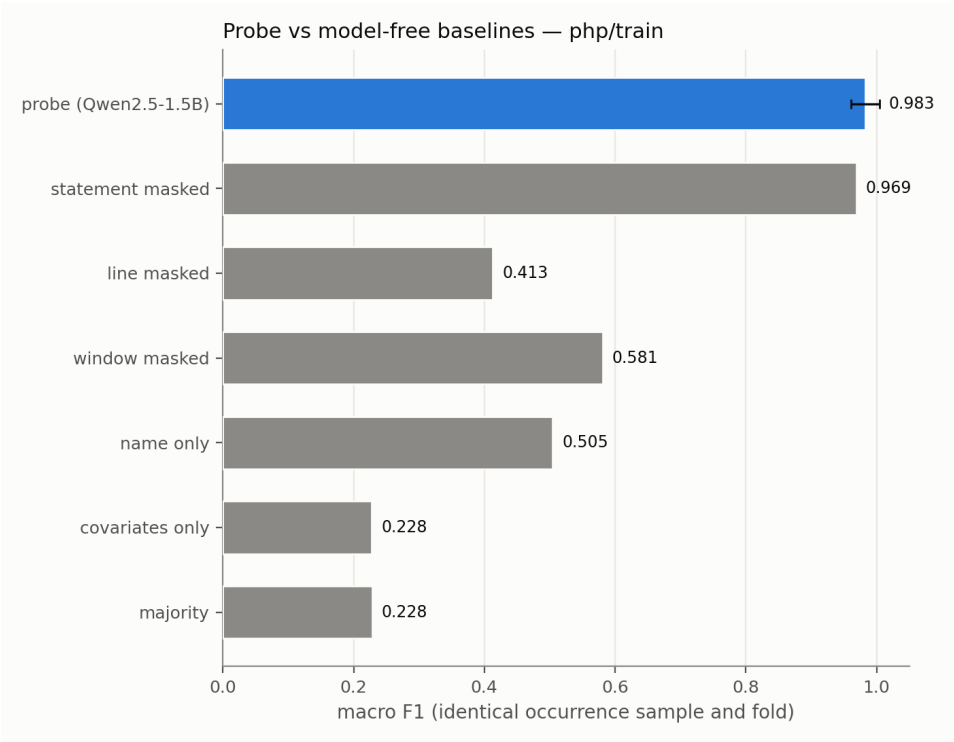

Figure 9: Boolean probe and model-free baselines for PHP, Qwen2.5-1.5B.

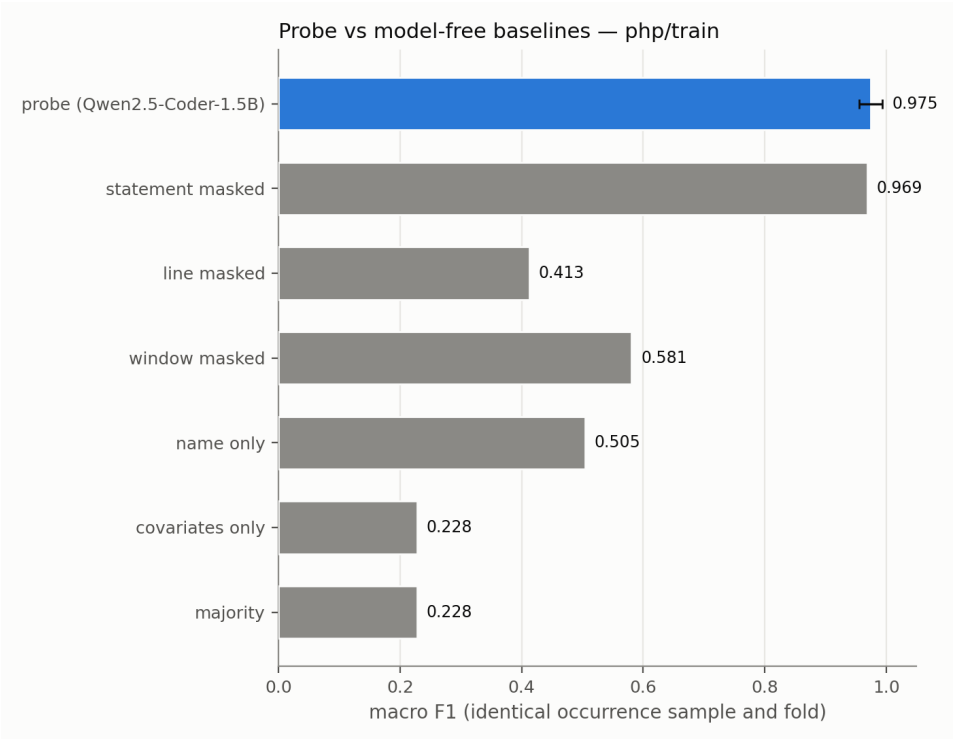

Figure 10: Boolean probe and model-free baselines for PHP, Qwen2.5-Coder-1.5B.

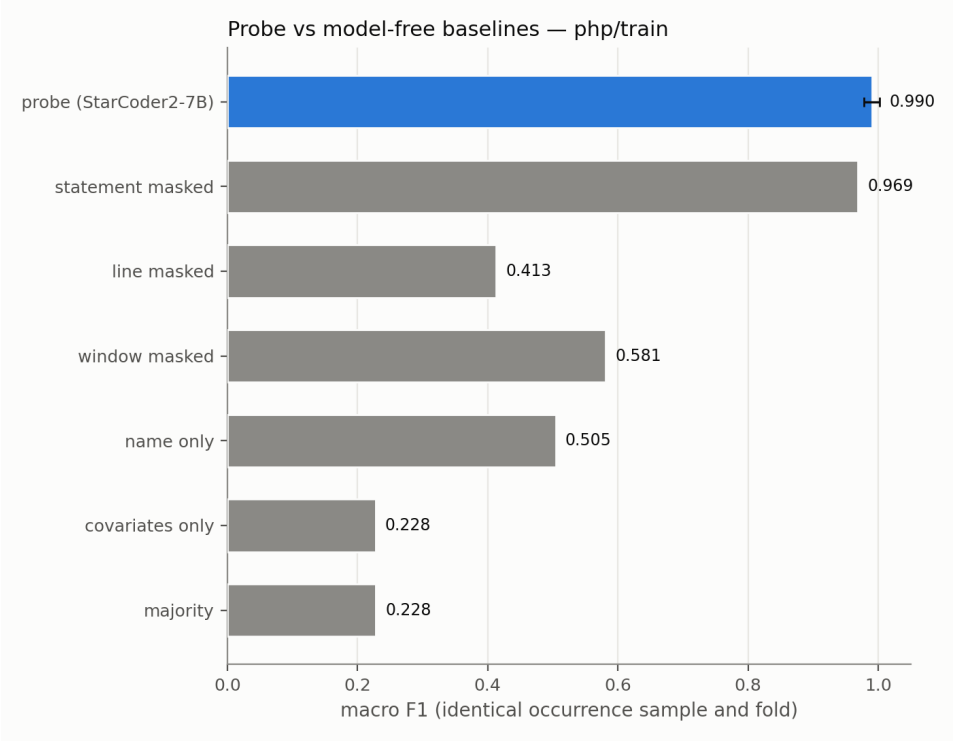

Figure 11: Boolean probe and model-free baselines for PHP, StarCoder2-7B.

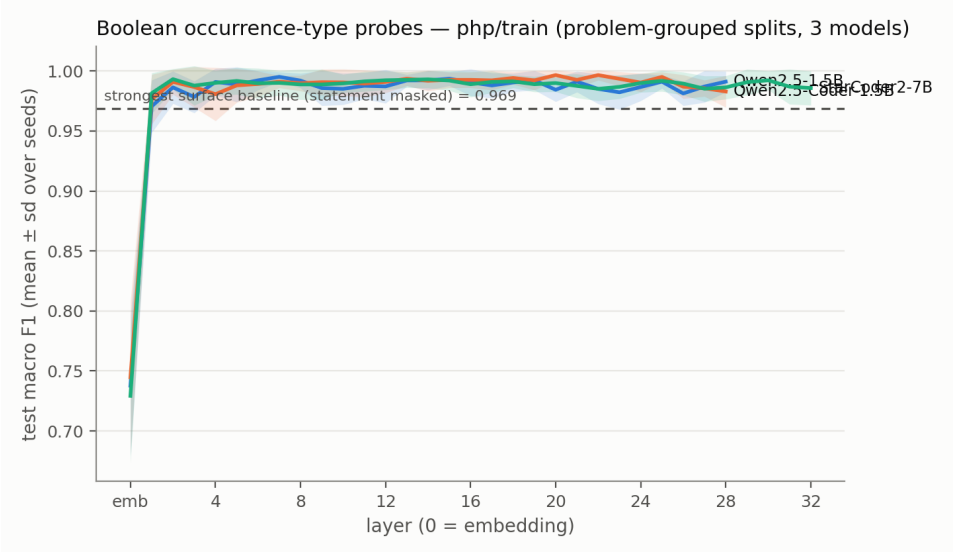

Figure 12: Boolean probe layer curves for PHP across the three models.

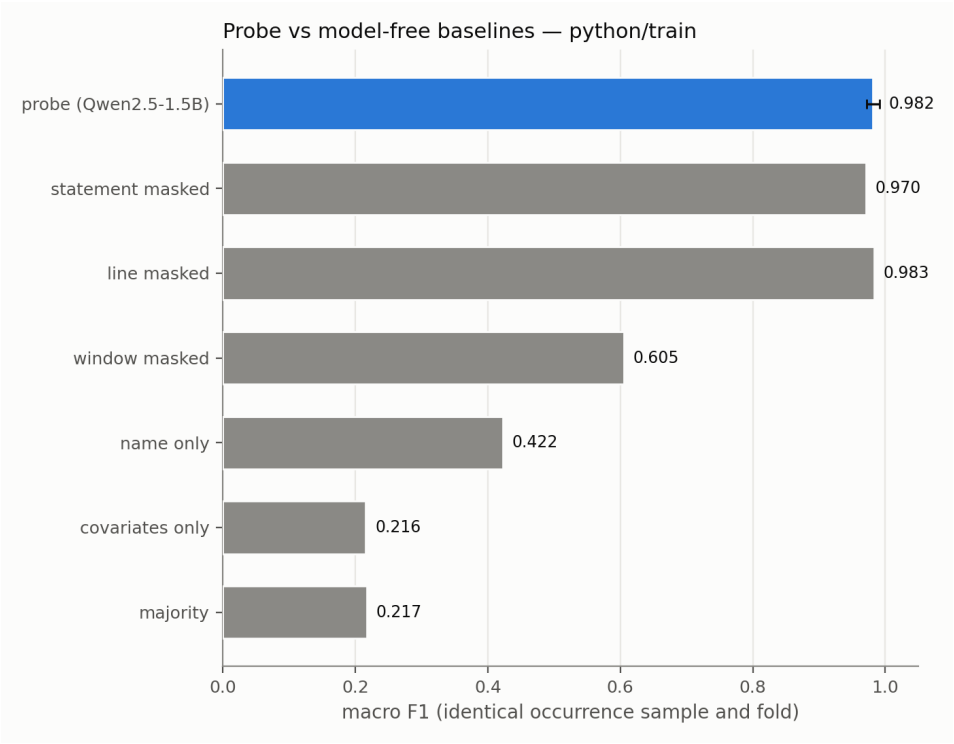

Figure 13: Boolean probe and model-free baselines for Python, Qwen2.5-1.5B.

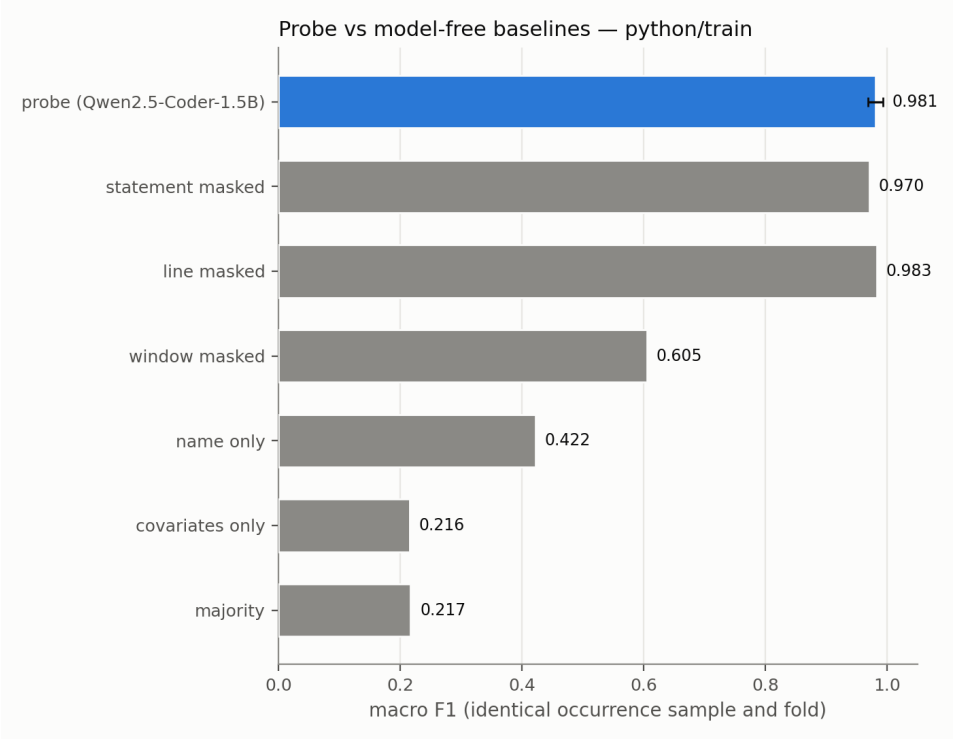

Figure 14: Boolean probe and model-free baselines for Python, Qwen2.5-Coder-1.5B.

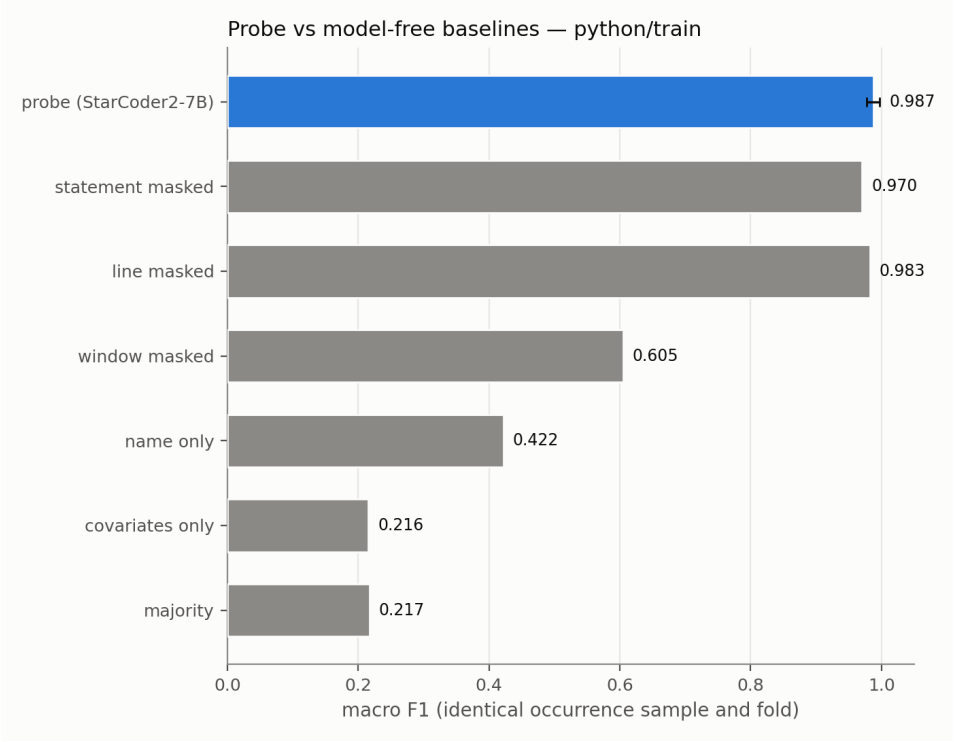

Figure 15: Boolean probe and model-free baselines for Python, StarCoder2-7B.

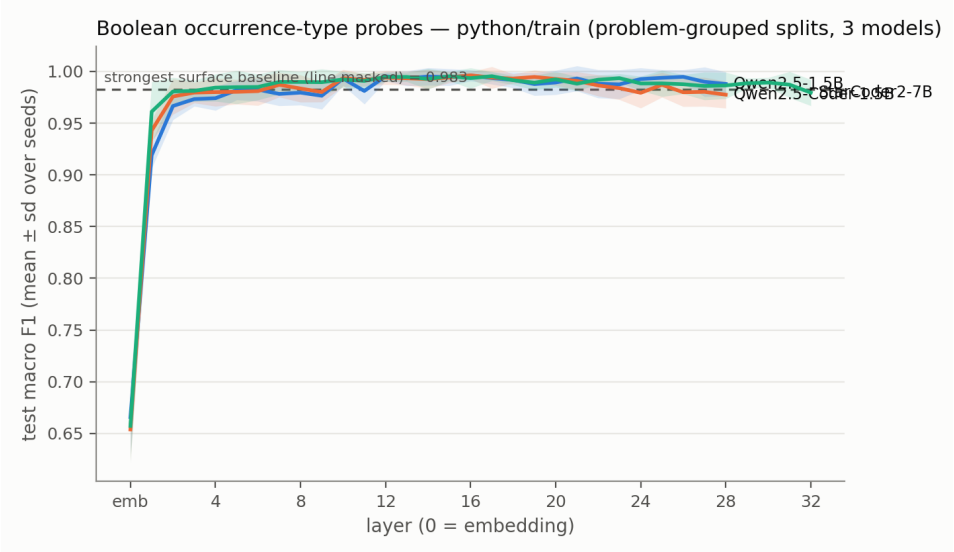

Figure 16: Boolean probe layer curves for Python across the three models.

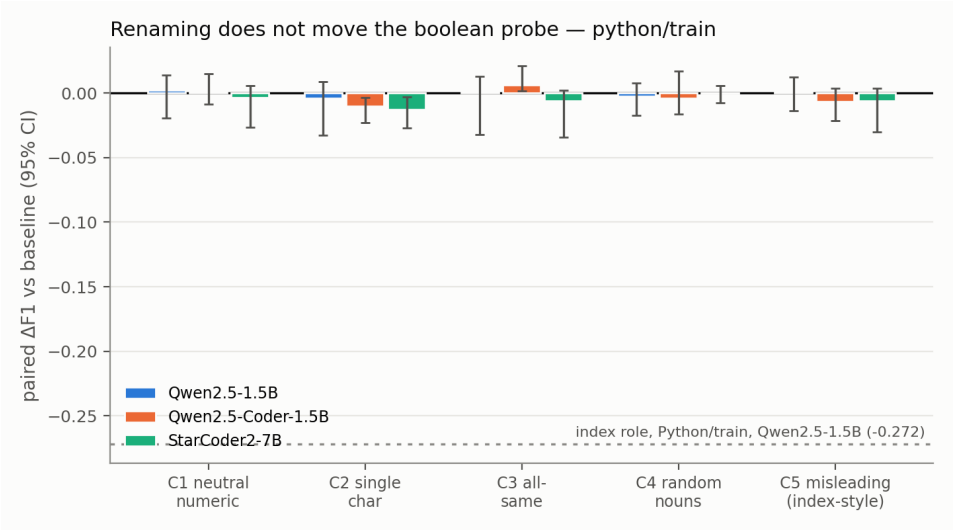

Figure 17: Problem-paired boolean refit-performance deltas on Python. No corresponding committed renaming run exists for other languages.

B Causal intervention details

Status. The iterator recovery exports predate the gate-specified class/structure protocol. They report descriptive recovery curves without clustered confidence intervals or matched placebo/random controls. The terminal value is one by construction, and C-language cross-language rows with zero pairs are missing data rather than null effects.

C Exploratory boolean interventions

Status and estimand. These boolean interventions are exploratory diagnostics rather than evidence for the main causal claim. The committed export contains layer-wise means but not the per-case arrays needed to recompute clustered intervals, and the runs do not use the specified gate of the class/structure experiment. The exporter defines effect as $|\mathrm{clean} - \mathrm{intervened}|$ and specificity as that effect divided by the corresponding random-position-control effect. Specificity can become large when its denominator is small. We therefore report both specificity at the largest-effect layer and the maximum specificity over layers rather than selecting only the more favorable one.

Full layer profiles. Each panel plots the clean readout, the intervention, and the available random-position, random-direction, or same-role controls. These figures preserve the layerwise information that is compressed by Tables 21–23.

Table 7: Complete class/structure perturbation results under the shared five-seed protocol. Each condition refits a probe and selects its own layer on validation data, so cross-condition deltas do not test a fixed direction.

model	strategy	layer	F1	95% CI	control F1	selectivity	Δ
Qwen2.5-1.5B	baseline	L18	0.979	[0.967, 0.987]	0.427	+0.552	—
Qwen2.5-1.5B	random_nouns	L20	0.986	[0.976, 0.992]	0.426	+0.560	+0.007
Qwen2.5-1.5B	single_chars	L11	0.957	[0.936, 0.969]	0.423	+0.534	-0.022
Qwen2.5-1.5B	all_same	L0	1.000	[1.000, 1.000]	0.440	+0.560	+0.021
Qwen2.5-1.5B	numeric_vars	L21	0.965	[0.952, 0.972]	0.422	+0.543	-0.014
Qwen2.5-1.5B	misleading_class_struct	L7	0.976	[0.962, 0.982]	0.416	+0.559	-0.004
Qwen2.5-Coder-1.5B	baseline	L8	0.982	[0.969, 0.989]	0.429	+0.553	—
Qwen2.5-Coder-1.5B	random_nouns	L19	0.978	[0.965, 0.985]	0.427	+0.551	-0.004
Qwen2.5-Coder-1.5B	single_chars	L7	0.956	[0.936, 0.967]	0.423	+0.533	-0.026
Qwen2.5-Coder-1.5B	all_same	L0	1.000	[1.000, 1.000]	0.440	+0.560	+0.018
Qwen2.5-Coder-1.5B	numeric_vars	L21	0.968	[0.955, 0.975]	0.424	+0.544	-0.014
Qwen2.5-Coder-1.5B	misleading_class_struct	L7	0.978	[0.967, 0.984]	0.415	+0.562	-0.004
StarCoder2-7B	baseline	L5	0.981	[0.967, 0.990]	0.478	+0.503	—
StarCoder2-7B	random_nouns	L6	0.976	[0.962, 0.983]	0.466	+0.510	-0.006
StarCoder2-7B	single_chars	L4	0.957	[0.921, 0.971]	0.475	+0.482	-0.025
StarCoder2-7B	all_same	L0	1.000	[1.000, 1.000]	0.445	+0.555	+0.019
StarCoder2-7B	numeric_vars	L23	0.968	[0.956, 0.975]	0.451	+0.517	-0.013
StarCoder2-7B	misleading_class_struct	L5	0.985	[0.976, 0.990]	0.467	+0.518	+0.004

Table 8: Class/structure cross-language results. Transfer uses the Python-selected layer; in-domain layers are selected separately. Only languages present in the committed exports are shown.

model	language	programs	in-domain layer	in-domain F1	transfer accuracy	transfer F1
Qwen2.5-1.5B	Python	400	L18	0.979	—	—
Qwen2.5-1.5B	C++	606	L10	0.991	0.993	0.933
Qwen2.5-1.5B	JavaScript	285	L5	0.987	0.998	0.957
Qwen2.5-1.5B	C	97	L8	0.986	0.984	0.886
Qwen2.5-Coder-1.5B	Python	400	L8	0.982	—	—
Qwen2.5-Coder-1.5B	C++	606	L8	0.988	0.991	0.918
Qwen2.5-Coder-1.5B	JavaScript	285	L6	0.985	0.996	0.913
Qwen2.5-Coder-1.5B	C	97	L20	0.980	0.983	0.885
StarCoder2-7B	Python	400	L5	0.981	—	—
StarCoder2-7B	C++	606	L5	0.992	0.996	0.965
StarCoder2-7B	JavaScript	285	L3	0.989	0.999	0.980
StarCoder2-7B	C	97	L5	0.986	0.987	0.906

Table 9: Layerwise class/structure robustness summaries. The F1 range is taken over all committed layers; maximum cosine similarity is measured against the baseline activation direction for each renamed condition.

model	strategy	minimum F1	maximum F1 (layer)	maximum cosine (layer)
Qwen2.5-1.5B	all_same	0.999	1.000 (L0)	0.105 (L11)
Qwen2.5-1.5B	baseline	0.806	0.983 (L18)	—
Qwen2.5-1.5B	misleading_class_struct	0.708	0.980 (L7)	0.311 (L15)
Qwen2.5-1.5B	numeric_vars	0.445	0.966 (L21)	0.329 (L22)
Qwen2.5-1.5B	random_nouns	0.529	0.989 (L15)	0.493 (L26)
Qwen2.5-1.5B	single_chars	0.530	0.965 (L7)	0.392 (L15)
Qwen2.5-Coder-1.5B	all_same	0.999	1.000 (L0)	0.126 (L28)
Qwen2.5-Coder-1.5B	baseline	0.807	0.985 (L7)	—
Qwen2.5-Coder-1.5B	misleading_class_struct	0.708	0.978 (L7)	0.324 (L7)
Qwen2.5-Coder-1.5B	numeric_vars	0.445	0.969 (L24)	0.352 (L22)
Qwen2.5-Coder-1.5B	random_nouns	0.530	0.983 (L23)	0.464 (L19)
Qwen2.5-Coder-1.5B	single_chars	0.530	0.959 (L7)	0.415 (L15)
StarCoder2-7B	all_same	0.999	1.000 (L0)	0.101 (L21)
StarCoder2-7B	baseline	0.763	0.984 (L6)	—
StarCoder2-7B	misleading_class_struct	0.717	0.985 (L3)	0.387 (L3)
StarCoder2-7B	numeric_vars	0.443	0.971 (L26)	0.369 (L22)
StarCoder2-7B	random_nouns	0.502	0.979 (L8)	0.511 (L3)
StarCoder2-7B	single_chars	0.529	0.962 (L3)	0.467 (L6)

Table 10: Complete iterator perturbation results. Each condition refits a probe and selects its own best layer. These single-run summaries do not carry the five-seed intervals used for the class/structure role.

model	strategy	best layer	accuracy	F1	Δ
Qwen2.5-Coder-1.5B	baseline	L6	0.998	0.988	—
Qwen2.5-Coder-1.5B	random_nouns	L9	0.996	0.974	-0.014
Qwen2.5-Coder-1.5B	single_chars	L9	0.985	0.923	-0.064
Qwen2.5-Coder-1.5B	all_same	L26	1.000	1.000	+0.012
Qwen2.5-Coder-1.5B	numeric_vars	L23	0.978	0.924	-0.064
Qwen2.5-Coder-1.5B	misleading_iterator	L3	1.000	1.000	+0.012
StarCoder2-7B	baseline	L5	0.999	0.990	—
StarCoder2-7B	random_nouns	L7	0.997	0.986	-0.005
StarCoder2-7B	single_chars	L5	0.993	0.961	-0.029
StarCoder2-7B	all_same	L4	1.000	1.000	+0.009
StarCoder2-7B	numeric_vars	L24	0.989	0.958	-0.032
StarCoder2-7B	misleading_iterator	L4	1.000	0.999	+0.009

Table 11: Complete iterator cross-language transfer results. Transfer columns use the Python-selected layer.

model	language	programs	in-domain layer	in-domain F1	transfer accuracy	transfer F1
Qwen2.5-Coder-1.5B	Python	256	L7	0.988	—	—
Qwen2.5-Coder-1.5B	Java	188	L12	0.996	0.991	0.950
Qwen2.5-Coder-1.5B	C++	191	L7	0.994	0.994	0.967
Qwen2.5-Coder-1.5B	C	191	L12	0.991	0.988	0.936
Qwen2.5-Coder-1.5B	C#	189	L13	0.998	0.992	0.957
Qwen2.5-Coder-1.5B	JavaScript	266	L12	0.992	0.995	0.978
Qwen2.5-Coder-1.5B	PHP	253	L8	0.995	0.997	0.983
StarCoder2-7B	Python	256	L10	0.991	—	—
StarCoder2-7B	Java	188	L3	0.997	0.989	0.929
StarCoder2-7B	C++	191	L5	0.997	0.991	0.941
StarCoder2-7B	C	191	L16	0.992	0.984	0.906
StarCoder2-7B	C#	189	L11	0.995	0.988	0.923
StarCoder2-7B	JavaScript	266	L16	0.998	0.986	0.931
StarCoder2-7B	PHP	253	L17	0.998	0.993	0.964

Table 12: Layerwise iterator robustness summaries. F1 ranges cover every committed layer; cosine similarity compares each perturbation direction with the baseline direction.

model	strategy	minimum F1	maximum F1 (layer)	maximum cosine (layer)
Qwen2.5-Coder-1.5B	all_same	1.000	1.000 (L26)	0.334 (L0)
Qwen2.5-Coder-1.5B	baseline	0.905	0.988 (L6)	—
Qwen2.5-Coder-1.5B	misleading_iterator	0.991	1.000 (L3)	0.187 (L15)
Qwen2.5-Coder-1.5B	numeric_vars	0.556	0.924 (L23)	0.253 (L19)
Qwen2.5-Coder-1.5B	random_nouns	0.633	0.974 (L9)	0.360 (L16)
Qwen2.5-Coder-1.5B	single_chars	0.664	0.923 (L9)	0.498 (L0)
StarCoder2-7B	all_same	0.999	1.000 (L4)	0.337 (L9)
StarCoder2-7B	baseline	0.898	0.990 (L5)	—
StarCoder2-7B	misleading_iterator	0.993	0.999 (L4)	0.265 (L6)
StarCoder2-7B	numeric_vars	0.552	0.958 (L24)	0.248 (L13)
StarCoder2-7B	random_nouns	0.613	0.986 (L7)	0.396 (L8)
StarCoder2-7B	single_chars	0.670	0.961 (L5)	0.451 (L0)

Table 13: Complete gate-specified class/structure patching summary in Qwen2.5-1.5B. The causal gate required mean effect at least 0.10, a positive interval, separation from controls, and recovery at least 0.05 in both directions.

layer	span	direction	control	n	mean effect	95% CI	recovery	minus random
0	query_name	denoise	target	288	+0.0000	[0.0000, 0.0000]	+0.0000	—
0	query_name	noise	target	288	+0.0000	[0.0000, 0.0000]	+0.0000	—
18	placebo	denoise	target	288	+0.0015	[-0.0010, 0.0043]	+0.0015	—
18	placebo	noise	target	288	+0.0006	[-0.0019, 0.0032]	+0.0006	—
18	query_name	denoise	random	288	-0.0107	[-0.0205, -0.0014]	-0.0110	—
18	query_name	denoise	same_source	288	+0.0004	[-0.0009, 0.0016]	+0.0004	—
18	query_name	denoise	target	288	+0.0087	[-0.0007, 0.0188]	+0.0089	+0.0194
18	query_name	noise	random	288	+0.0096	[0.0023, 0.0164]	+0.0099	—
18	query_name	noise	same_source	288	-0.0005	[-0.0018, 0.0006]	-0.0006	—
18	query_name	noise	target	288	+0.0199	[0.0109, 0.0296]	+0.0204	+0.0103

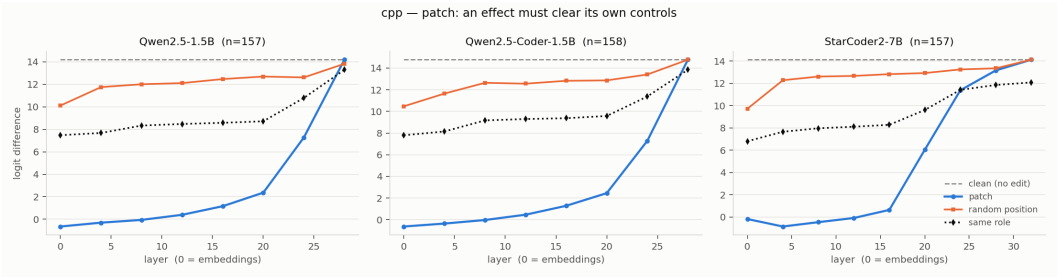

Figure 18: Boolean patch layer profiles for C++ across all three models and available controls.

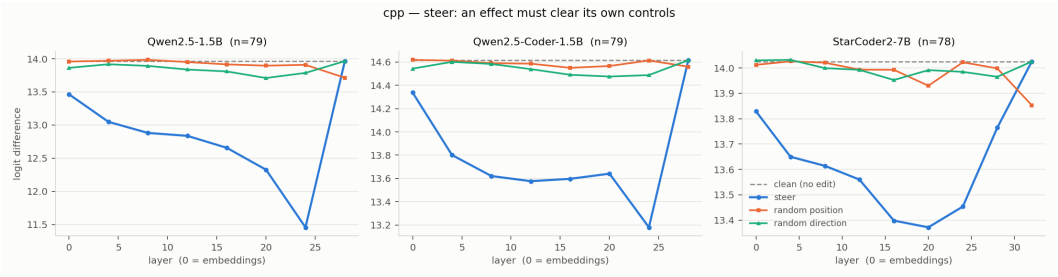

Figure 19: Boolean steer layer profiles for C++ across all three models and available controls.

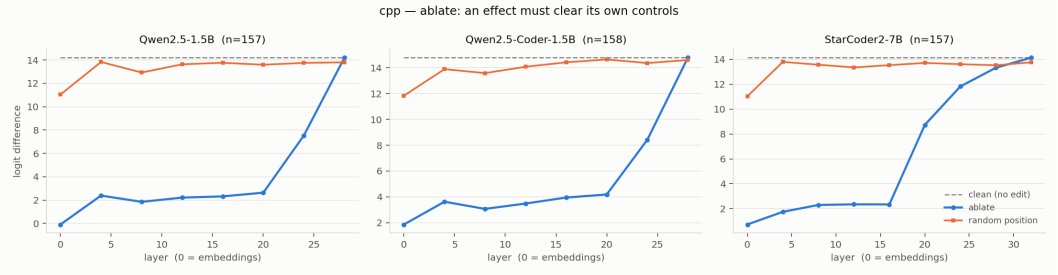

Figure 20: Boolean ablate layer profiles for C++ across all three models and available controls.

Table 14: Aggregate probe-link diagnostic for the class/structure evaluation pairs. Absolute values are reported because pair orientation is arbitrary. Large probe movement alongside small behavioral movement is descriptive of the failed causal transfer, not evidence that probe movement causes behavior.

quantity	mean absolute	median absolute
probe movement	15.786	16.302
behavioral movement	0.038	0.031

Table 15: Class/structure gate and readout audit. Both prompt classes favor `True`; the pairwise gap remains separable, but the binary readout is not calibrated to distinguish function prompts.

diagnostic	estimate	95% CI	gate
class logit difference	2.752	[2.624, 2.898]	pass
function logit difference	1.778	[1.652, 1.909]	fail
class-minus-function gap	0.974	[0.908, 1.035]	pass
pair-gap accuracy	0.986	[0.962, 0.993]	pass
query/declaration probe AUC	0.943/0.905	—	pass

Table 16: Patching audit trail. Completeness establishes that the scheduled Qwen evaluation finished; it does not turn the failed causal gate into positive evidence. The controller stopped before Coder and StarCoder2 after the Qwen null.

audit item	value
behavior rows present/expected	1152/1152
primary rows present/expected	4032/4032
baseline probe-gap/behavior Spearman	0.164 [0.023, 0.286]
controller terminal state	`stopped_qwen_null`

Table 17: Exploratory iterator activation-patching recovery for Qwen2.5-Coder-1.5B, within-language perturbations.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5	L6
random_nouns	L6	150	0.999/0.138	0.991	0.930	0.994	0.996	0.998	1.000
single_chars	L6	150	0.996/0.614	0.860	0.799	0.935	0.963	0.983	1.000
all_same	L6	150	0.999/0.142	0.808	0.900	0.961	0.986	0.998	1.000
numeric_vars	L6	150	0.999/0.057	0.966	0.961	0.991	0.997	0.999	1.000
misleading_iterator	L6	150	0.999/0.286	0.820	0.835	0.916	0.946	0.983	1.000

Table 18: Exploratory iterator activation-patching recovery for Qwen2.5-Coder-1.5B, cross-language transfer.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5	L6	L7
Java	L7	10	0.989/0.709	-0.300	-0.146	0.616	0.867	0.845	0.940	1.000
C++	L7	11	0.979/0.769	-0.066	0.025	0.785	0.869	0.869	0.904	1.000
C	L7	0	0.000/0.000	—	—	—	—	—	—	—
C#	L7	9	0.981/0.629	0.307	0.353	0.772	0.867	0.903	0.989	1.000
JavaScript	L7	13	0.983/0.766	0.708	0.678	0.968	0.943	0.953	0.983	1.000
PHP	L7	7	0.982/0.748	-0.211	-0.397	0.614	0.760	0.934	1.027	1.000

Table 19: Exploratory iterator activation-patching recovery for StarCoder2-7B, within-language perturbations.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5
random_nouns	L5	150	0.998/0.569	0.931	0.766	0.966	0.951	1.000
single_chars	L5	150	0.994/0.686	0.636	0.006	0.823	0.759	1.000
all_same	L5	150	0.999/0.249	0.342	0.772	0.946	0.979	1.000
numeric_vars	L5	150	0.998/0.530	0.889	0.811	0.953	0.955	1.000
misleading_iterator	L5	150	0.997/0.628	0.777	0.104	0.921	0.843	1.000

Table 20: Exploratory iterator activation-patching recovery for StarCoder2-7B, cross-language transfer.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
Java	L10	150	0.996/0.750	0.254	0.634	0.829	0.913	0.957	0.964	0.979	0.988	0.994	1.000
C++	L10	150	0.996/0.771	0.337	0.756	0.892	0.938	0.966	0.969	0.979	0.988	0.995	1.000
C	L10	0	0.000/0.000	—	—	—	—	—	—	—	—	—	—
C#	L10	150	0.996/0.740	0.294	0.692	0.877	0.942	0.969	0.974	0.984	0.991	0.996	1.000
JavaScript	L10	150	0.996/0.716	0.311	0.743	0.880	0.939	0.972	0.976	0.983	0.990	0.996	1.000
PHP	L10	73	0.993/0.808	0.491	0.796	0.938	0.964	0.979	0.973	0.983	0.994	0.999	1.000

Table 21: Exploratory boolean patch diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

language	model	n	peak layer	peak effect	spec. at peak	best spec. (layer)
C++	Qwen2.5-1.5B	157	L0	14.845	3.6×	7.9× (L20)
C++	Qwen2.5-Coder-1.5B	158	L0	15.414	3.6×	6.9× (L8)
C++	StarCoder2-7B	157	L4	14.982	8.1×	10.4× (L16)
Java	Qwen2.5-1.5B	155	L8	11.607	6.1×	7.7× (L12)
Java	Qwen2.5-Coder-1.5B	155	L8	12.642	5.9×	7.7× (L12)
Java	StarCoder2-7B	155	L4	13.900	7.5×	8.6× (L12)
JavaScript	Qwen2.5-1.5B	58	L8	12.755	6.9×	6.9× (L8)
JavaScript	Qwen2.5-Coder-1.5B	62	L8	12.717	6.6×	7.4× (L12)
JavaScript	StarCoder2-7B	129	L4	12.137	8.1×	8.2× (L16)
PHP	Qwen2.5-1.5B	15	L4	11.698	7.2×	7.2× (L4)
PHP	Qwen2.5-Coder-1.5B	16	L4	11.425	4.7×	7.4× (L0)
PHP	StarCoder2-7B	17	L4	11.545	6.3×	9.3× (L16)
Python	Qwen2.5-1.5B	176	L8	12.948	6.0×	7.2× (L16)
Python	Qwen2.5-Coder-1.5B	176	L8	13.275	6.7×	7.8× (L16)
Python	StarCoder2-7B	176	L4	13.592	7.9×	9.7× (L12)

Table 22: Exploratory boolean steer diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

language	model	n	peak layer	peak effect	spec. at peak	best spec. (layer)
C++	Qwen2.5-1.5B	79	L24	2.508	45.0×	169.3× (L4)
C++	Qwen2.5-Coder-1.5B	79	L24	1.436	602.0×	602.0× (L24)
C++	StarCoder2-7B	78	L20	0.653	6.9×	345.4× (L4)
Java	Qwen2.5-1.5B	78	L24	2.486	72.2×	826.9× (L8)
Java	Qwen2.5-Coder-1.5B	78	L24	1.712	72.0×	179.9× (L0)
Java	StarCoder2-7B	78	L24	0.555	244.0×	244.0× (L24)
JavaScript	Qwen2.5-1.5B	31	L24	1.763	24.5×	76.8× (L8)
JavaScript	Qwen2.5-Coder-1.5B	31	L24	1.147	65.3×	147.6× (L4)
JavaScript	StarCoder2-7B	65	L16	0.591	11.2×	61.0× (L28)
PHP	Qwen2.5-1.5B	7	L24	2.759	190.2×	190.2× (L24)
PHP	Qwen2.5-Coder-1.5B	8	L24	0.898	11.5×	132.7× (L8)
PHP	StarCoder2-7B	9	L20	0.991	8.7×	99.9× (L4)
Python	Qwen2.5-1.5B	88	L24	2.175	209.1×	794.6× (L16)
Python	Qwen2.5-Coder-1.5B	88	L24	1.231	1027.1×	1027.1× (L24)
Python	StarCoder2-7B	88	L20	0.721	8.6×	55.1× (L4)

Table 23: Exploratory boolean ablate diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

language	model	n	peak layer	peak effect	spec. at peak	best spec. (layer)
C++	Qwen2.5-1.5B	157	L0	14.286	4.5×	31.9× (L4)
C++	Qwen2.5-Coder-1.5B	158	L0	12.927	4.4×	76.2× (L20)
C++	StarCoder2-7B	157	L0	13.405	4.3×	36.5× (L4)
Java	Qwen2.5-1.5B	155	L16	9.720	28.5×	86.0× (L12)
Java	Qwen2.5-Coder-1.5B	155	L4	9.331	12.3×	62.4× (L12)
Java	StarCoder2-7B	155	L4	11.190	27.0×	27.0× (L4)
JavaScript	Qwen2.5-1.5B	58	L16	10.852	26.7×	26.7× (L16)
JavaScript	Qwen2.5-Coder-1.5B	62	L16	9.868	28.1×	116.4× (L24)
JavaScript	StarCoder2-7B	129	L4	10.970	27.3×	27.3× (L4)
PHP	Qwen2.5-1.5B	15	L16	15.698	15.0×	102.4× (L20)
PHP	Qwen2.5-Coder-1.5B	16	L16	13.402	26.5×	79.2× (L20)
PHP	StarCoder2-7B	17	L4	14.260	19.4×	19.7× (L8)
Python	Qwen2.5-1.5B	176	L0	12.144	5.8×	36.7× (L24)
Python	Qwen2.5-Coder-1.5B	176	L0	11.286	6.8×	33.1× (L16)
Python	StarCoder2-7B	176	L4	11.818	26.5×	26.5× (L4)

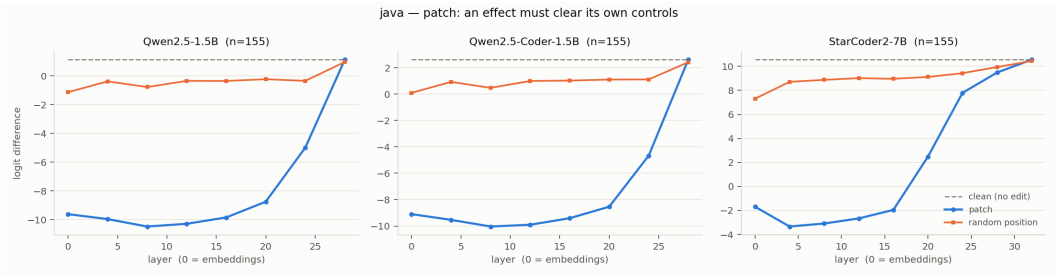

Figure 21: Boolean patch layer profiles for Java across all three models and available controls.

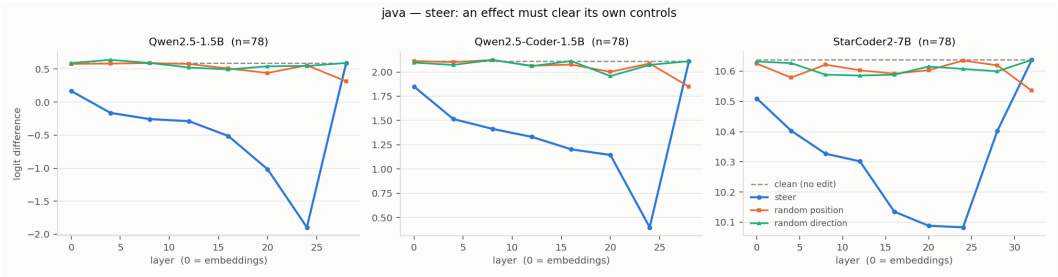

Figure 22: Boolean steer layer profiles for Java across all three models and available controls.

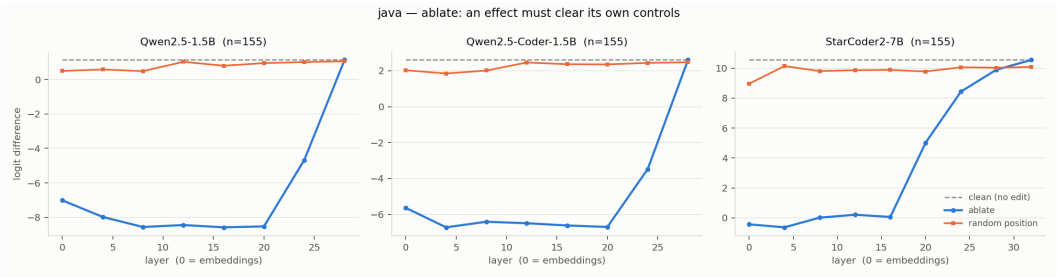

Figure 23: Boolean ablate layer profiles for Java across all three models and available controls.

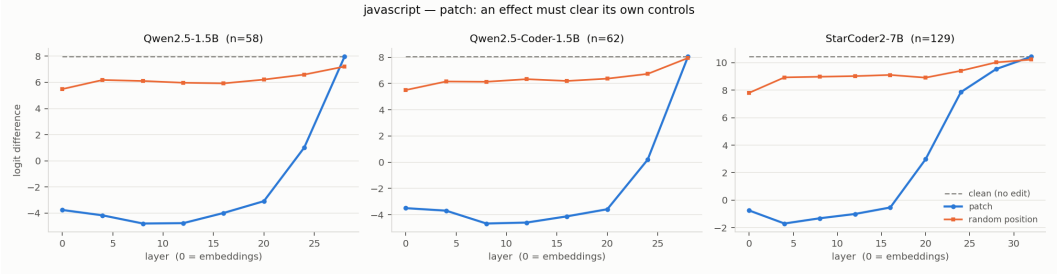

Figure 24: Boolean patch layer profiles for JavaScript across all three models and available controls.

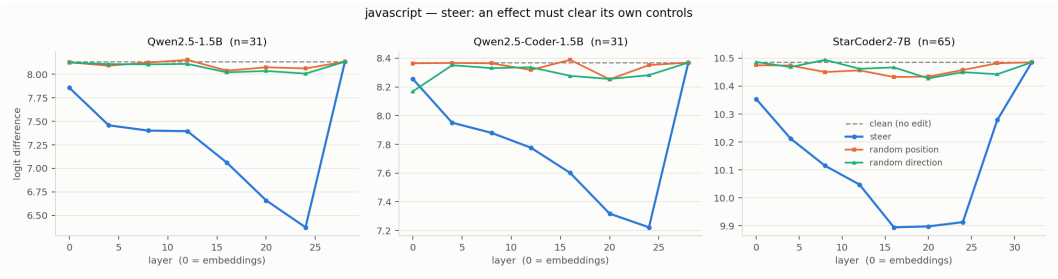

Figure 25: Boolean steer layer profiles for JavaScript across all three models and available controls.

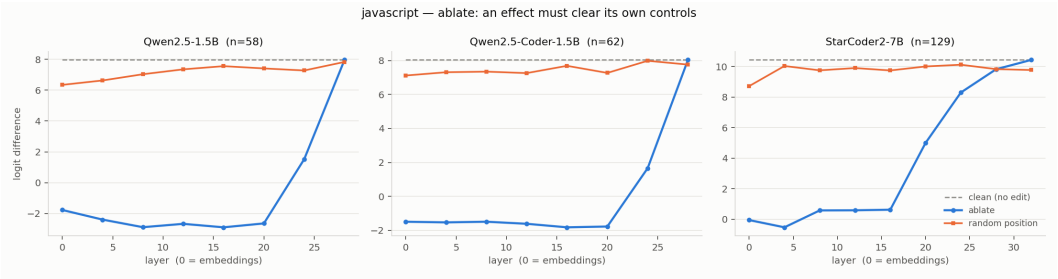

Figure 26: Boolean ablate layer profiles for JavaScript across all three models and available controls.

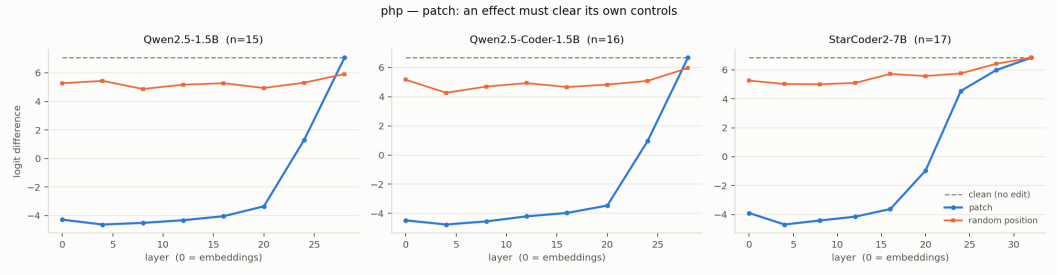

Figure 27: Boolean patch layer profiles for PHP across all three models and available controls.

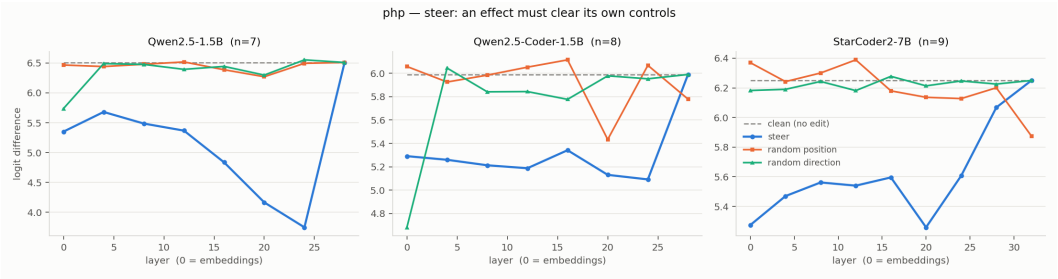

Figure 28: Boolean steer layer profiles for PHP across all three models and available controls.

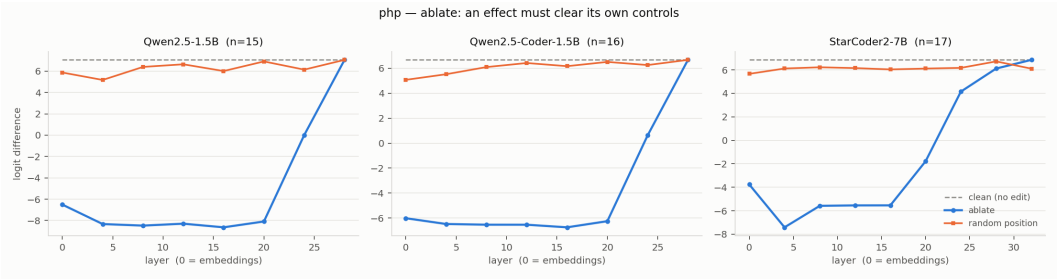

Figure 29: Boolean ablate layer profiles for PHP across all three models and available controls.

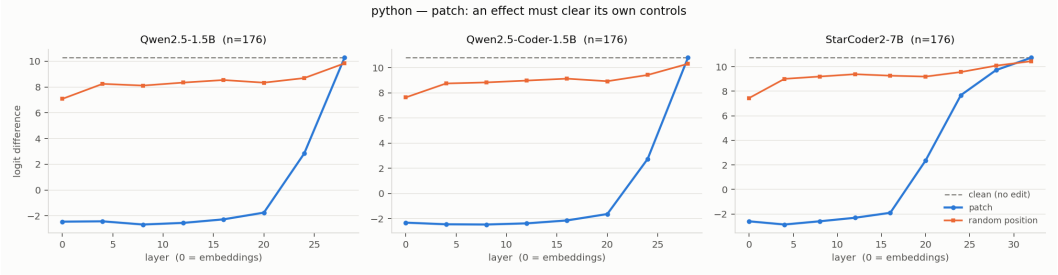

Figure 30: Boolean patch layer profiles for Python across all three models and available controls.

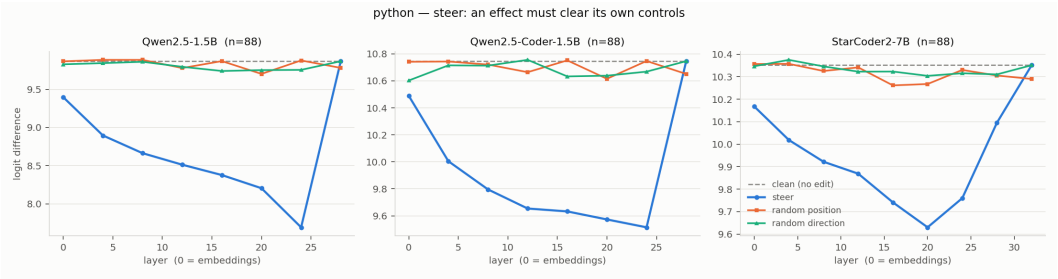

Figure 31: Boolean steer layer profiles for Python across all three models and available controls.

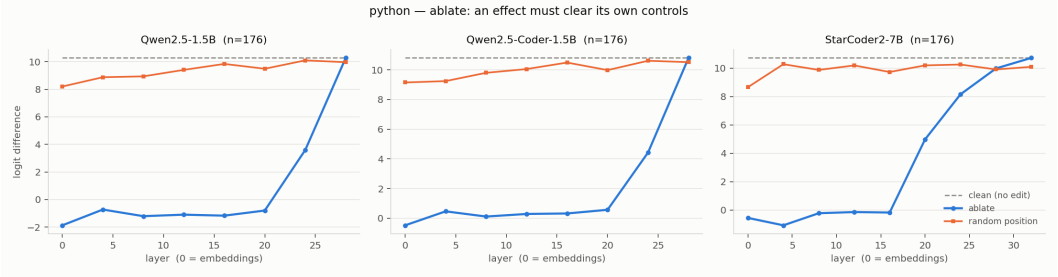

Figure 32: Boolean ablate layer profiles for Python across all three models and available controls.

D Committed role-analysis figures

Figure provenance. These are committed analysis figures. They supplement the generated tables but do not add uncertainty estimates where the underlying export lacks them.

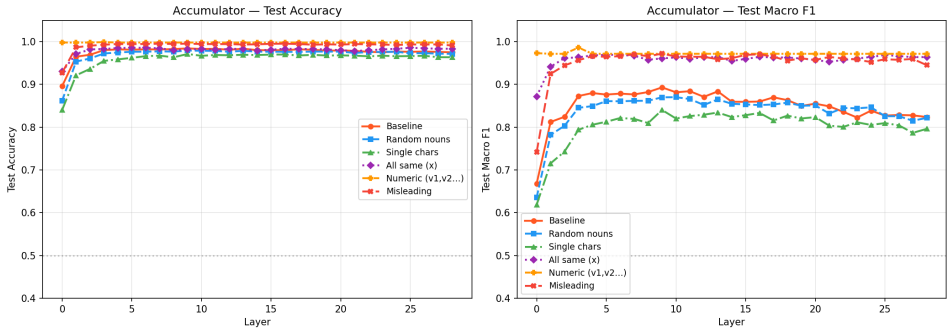

Figure 33: Accumulator probe accuracy and macro-F1 across layers and all six identifier conditions.

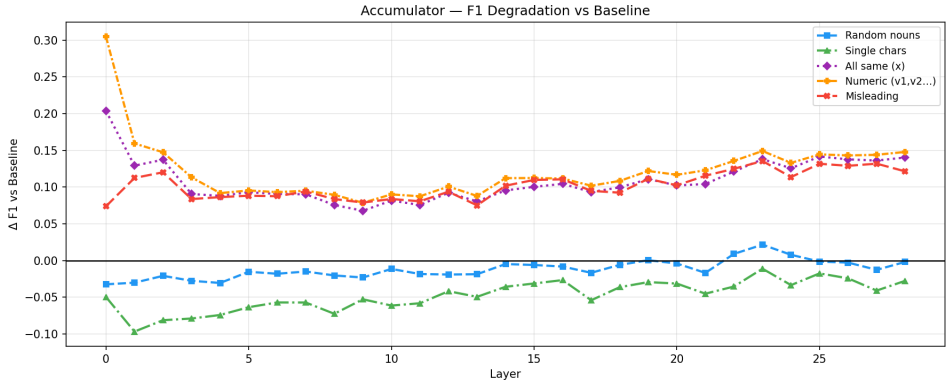

Figure 34: Accumulator macro-F1 changes relative to baseline naming.

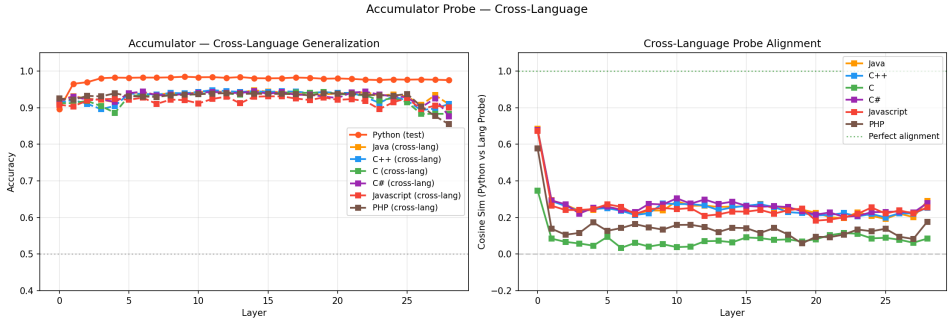

Figure 35: Accumulator in-domain and Python-trained cross-language results for the available compiled languages.

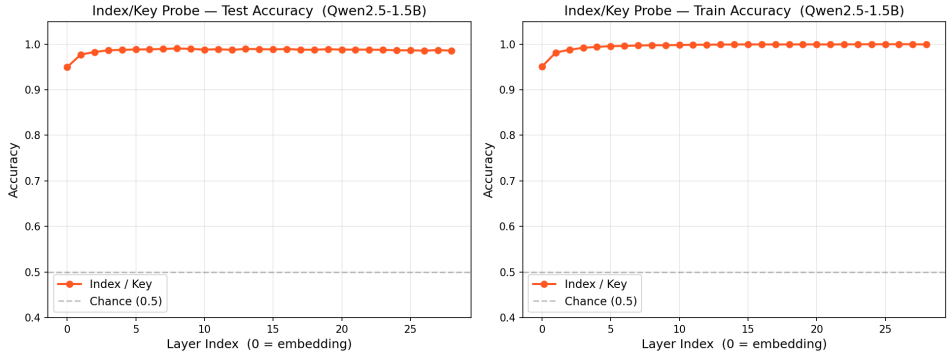

Figure 36: Index/key probe accuracy and macro-F1 across layers.

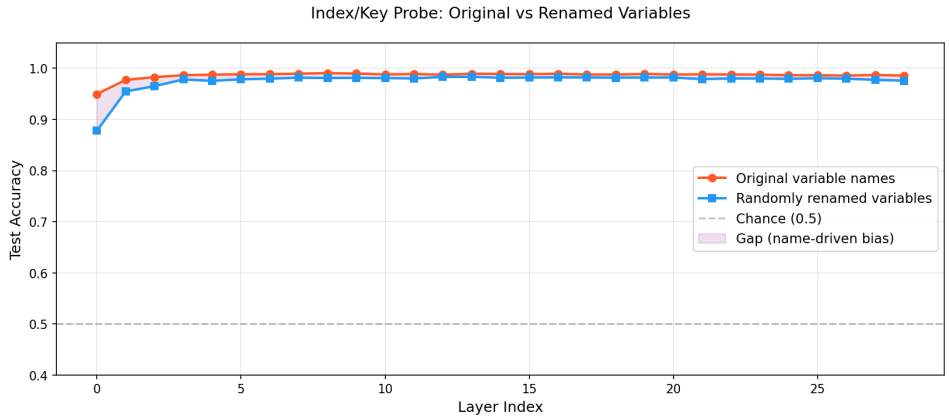

Figure 37: Index/key probe comparison under original and misleading names.

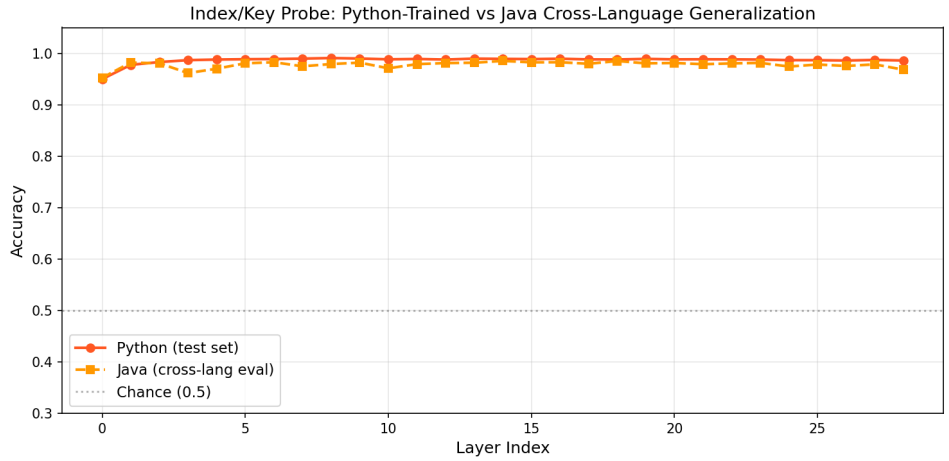

Figure 38: Index/key in-domain and Python-trained cross-language results.

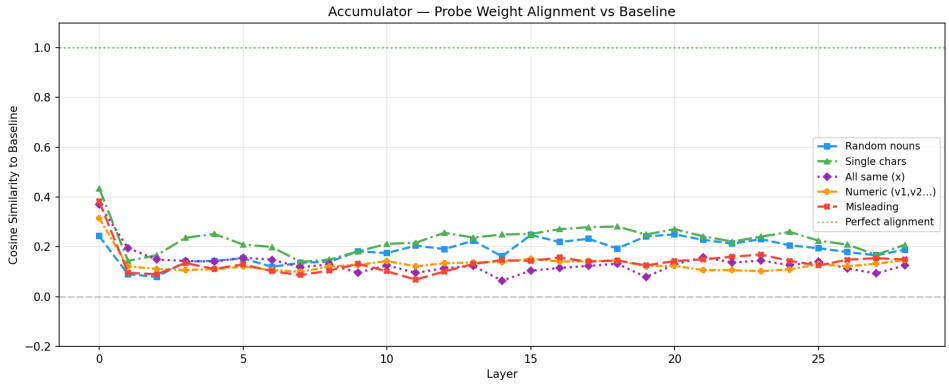

Figure 39: Legacy accumulator probe-direction cosine similarity under the identifier interventions.

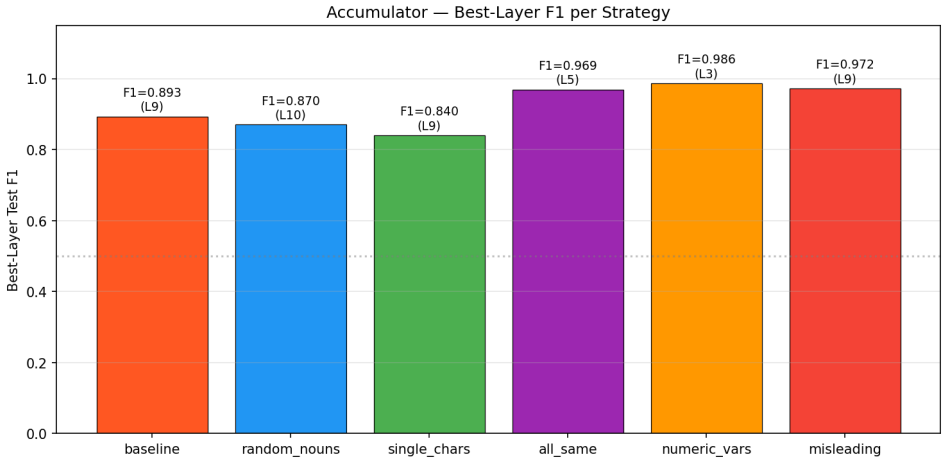

Figure 40: Legacy accumulator best-layer macro-F1 by identifier condition.

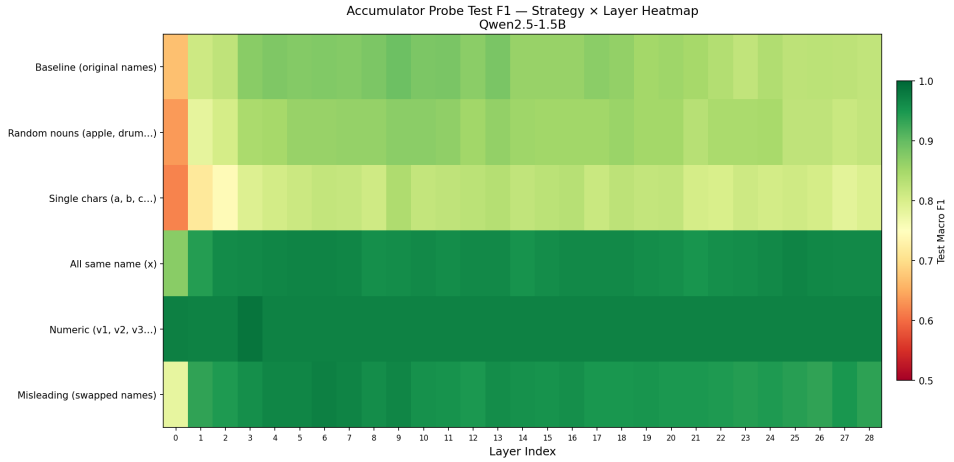

Figure 41: Legacy accumulator transfer heatmap over the available languages.

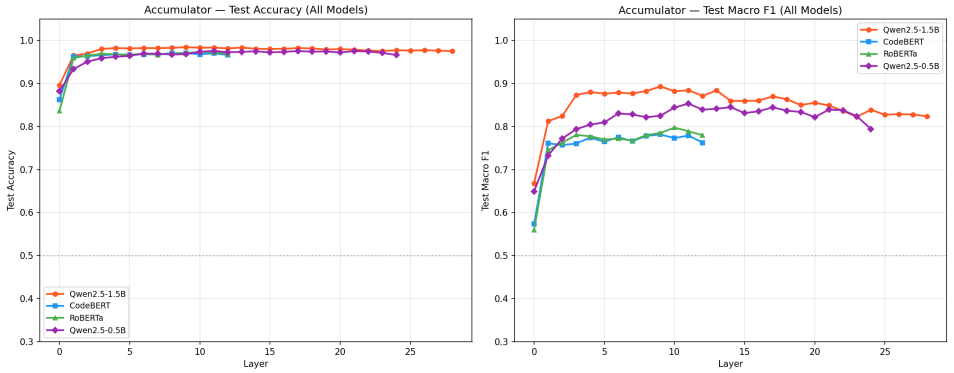

Figure 42: Legacy accumulator multi-model comparison. This predates the shared modern protocol and is descriptive only.

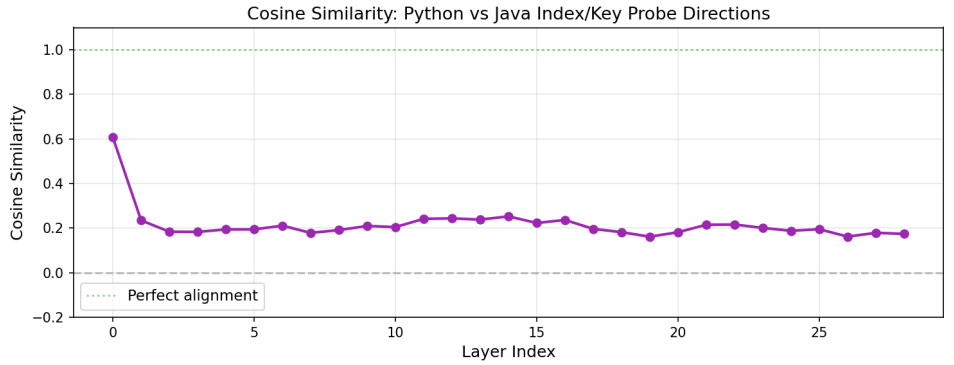

Figure 43: Legacy index/key probe-direction cosine similarity across layers.

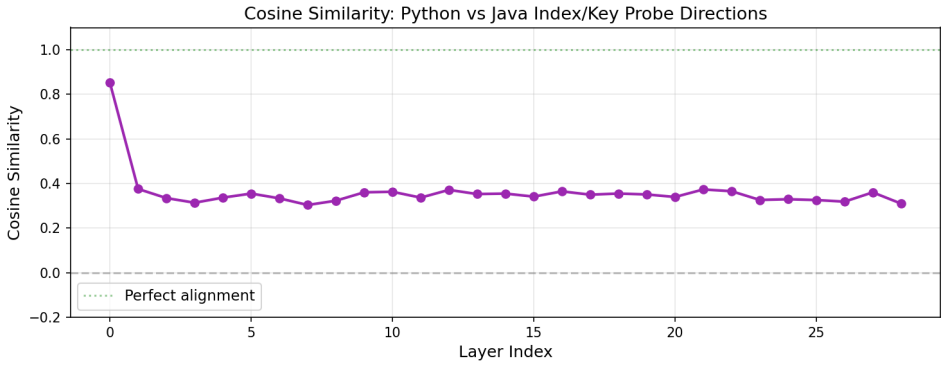

Figure 44: Index/key cosine-similarity sensitivity run using the smaller 100-example configuration.

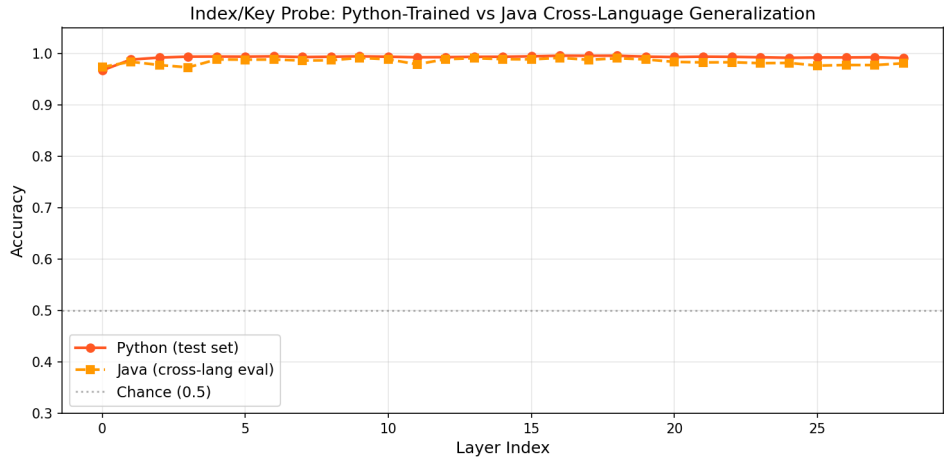

Figure 45: Index/key cross-language sensitivity run using the smaller 100-example configuration.

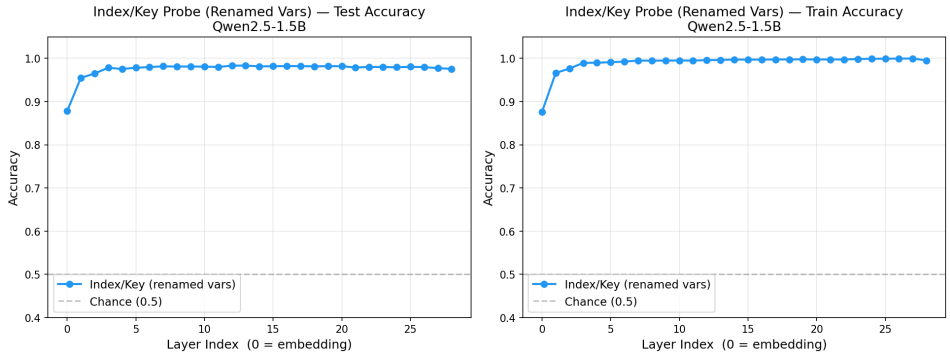

Figure 46: Legacy index/key layer curves under the full renamed condition.

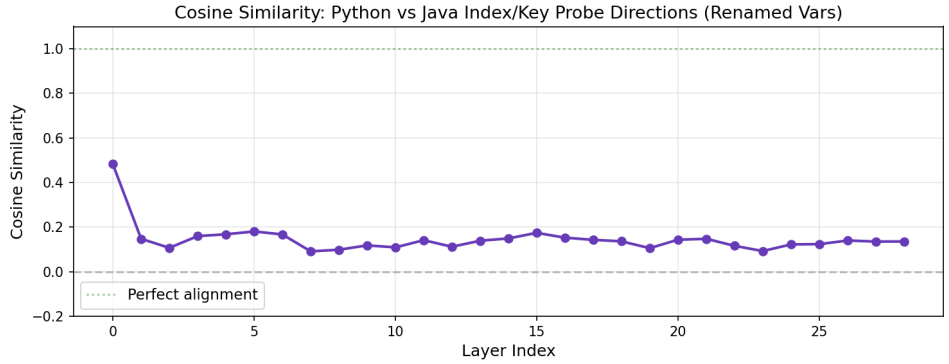

Figure 47: Legacy index/key cosine similarity under renaming.

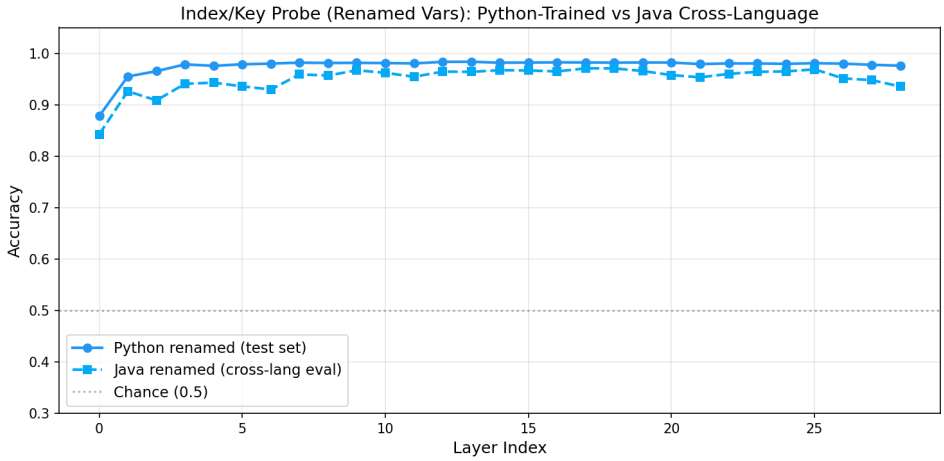

Figure 48: Legacy index/key cross-language transfer under renaming.

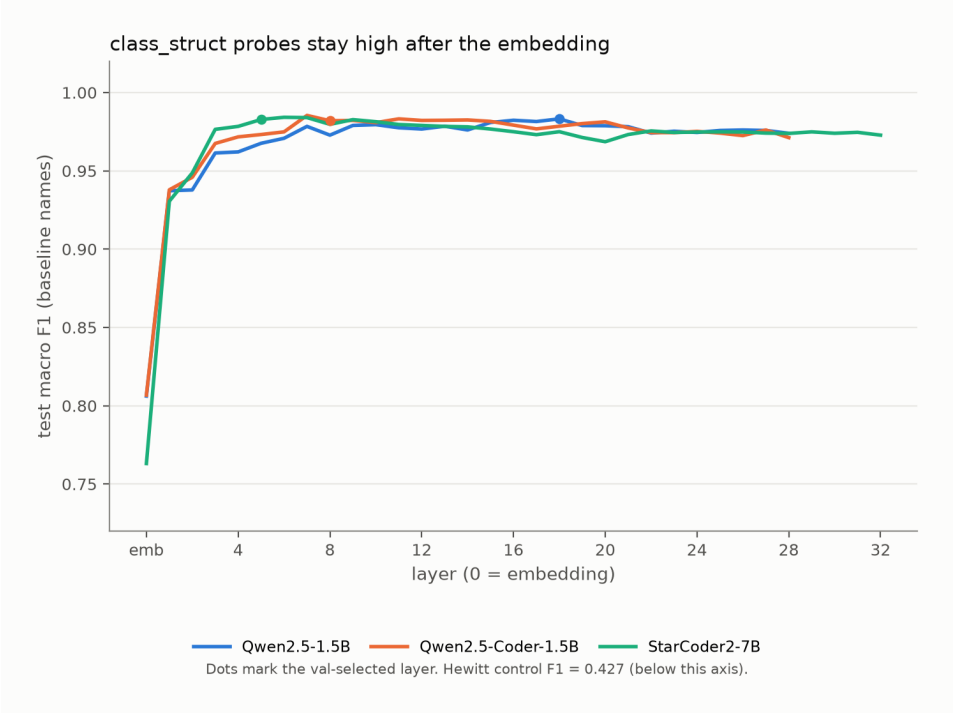

Figure 49: Class/structure probe macro-F1 across layers and perturbations.

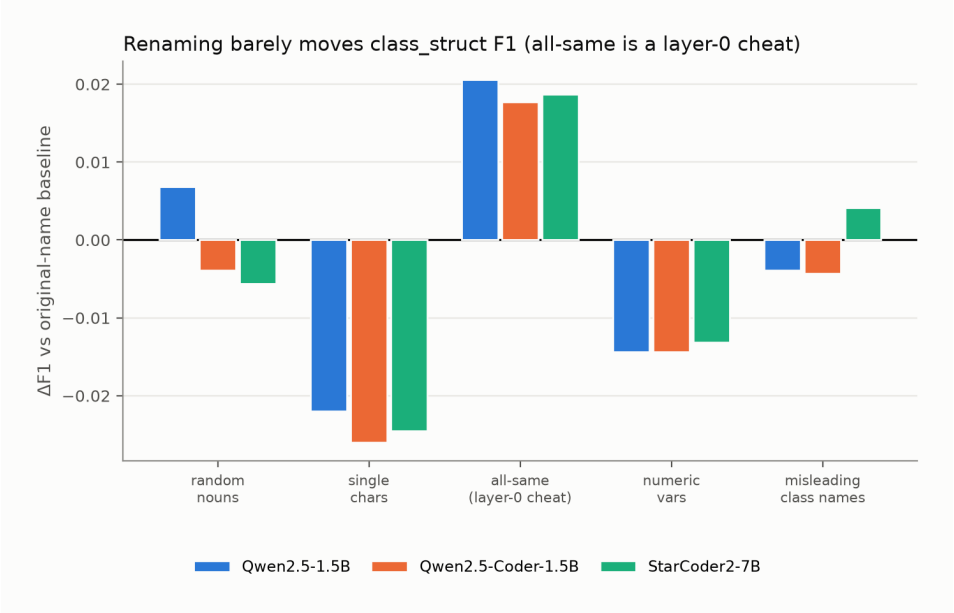

Figure 50: Class/structure perturbation deltas relative to baseline.

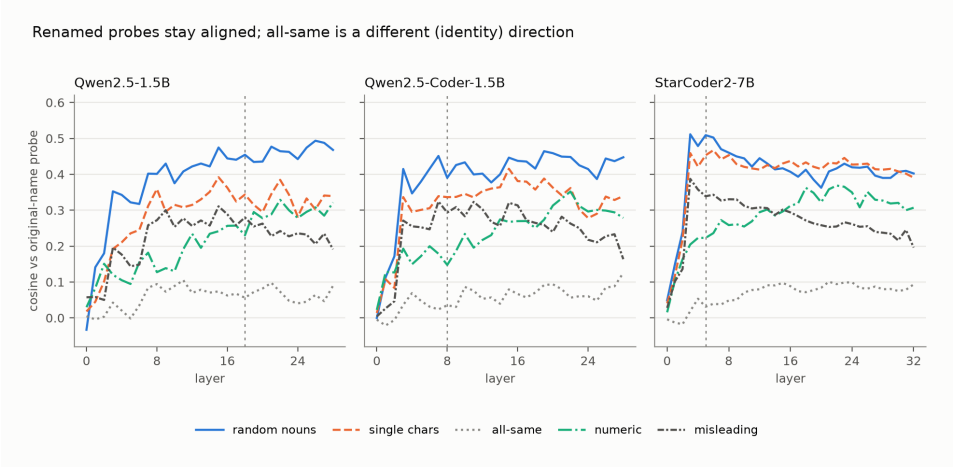

Figure 51: Class/structure cosine similarity to the baseline direction.

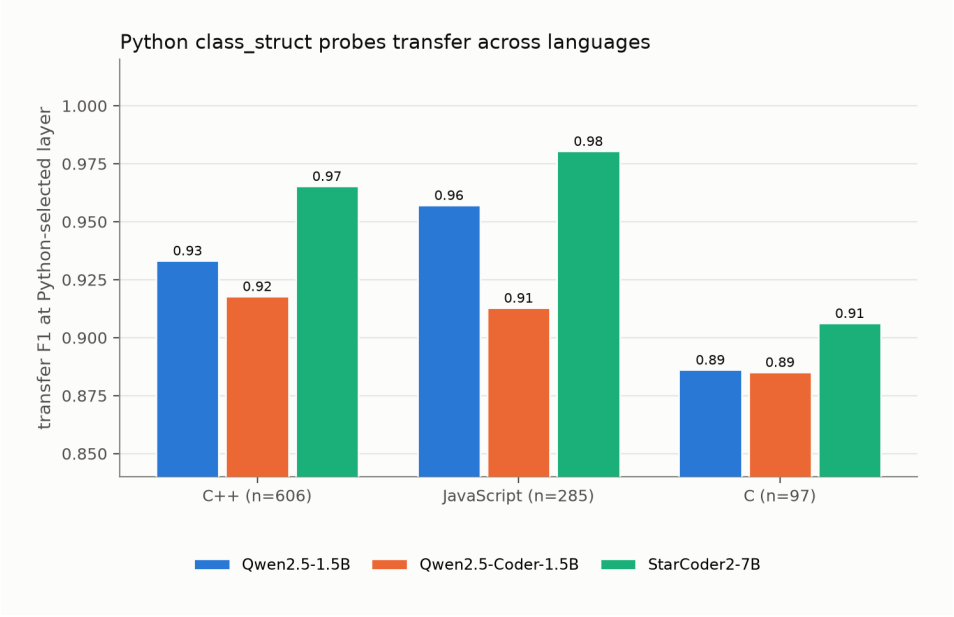

Figure 52: Available class/structure cross-language transfer results.

E Explicit coverage holes

- Accumulator and index/key results use legacy exports; no committed problem-grouped, five-seed CSV reproduces those roles under the modern protocol.
- Iterator results are available for Qwen2.5-Coder-1.5B and StarCoder2-7B, but not Qwen2.5-1.5B.
- Class/structure transfer exports include Python, C++, JavaScript, and C, but not Java, C#, or PHP.
- Boolean probe exports include Python, Java, JavaScript, and PHP; C, C#, and C++ are absent, and renaming is available only for Python.
- Gate-specified class/structure patching stopped after the Qwen2.5-1.5B null; Coder, StarCoder2, and the all-layer core sweep were not attempted.
- No matched model-free surface comparator is committed for the class/structure role.

F Reproducibility

The appendix is generated from committed boolean, iterator, class-reference, and patching artifacts by `scripts/make_interp_appendix.py`. Modern probe conditions use frozen problem hashes, validation-selected layers, and five seeds where stated; iterator summaries are single-run. The class-reference intervention records model and dataset revisions, prompt and configuration hashes, all $4{,}032$ primary rows, and clustered intervals. A regression test checks source traces for the headline numbers and rejects LP4FM-only conditions in this manuscript.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction distinguish decodability, lexical robustness, surface irreducibility, and causal use. They state that the study includes three models, five roles, and seven languages collectively without claiming a complete factorial design.

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   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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